Project 1

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PBHLT 7101: Data Management, Analytics and Visualization
March 3, 2026

1. Introduction

The purpose of this analysis was to practice preparing a dataset for an exploratory data analysis and then conducting an exploratory data analysis. The working dataset used for preparation and analysis was derived from the 2012 US Army Anthropometric Survey (ANSUR II), a publicly available dataset collected by the Natick Soldier Research, Development and Engineering Center (NSRDEC) between October 2010 and April 2012.[1] The ANSUR II data were collected from subjects in the Army Reserve, Army National Guard, and Active Duty Army located at various Army installations.[1] The complete ANSUR II database includes 93 directly measured anthropometric measures and 15 demographic/administrative variables and is divided into separate male (n=4,082) and female (n=1,986) databases.[1]

For practice and learning purposes, Dr. Vanderslice selected a subset of variables from the ANSUR II database and modified various aspects of the data to create the working dataset used for this analysis. The male and female datasets were also combined into this working dataset. The working dataset initially contained 7,031 observations, indicating that duplicate observations were introduced by Dr. Vanderslice. The working dataset includes demographic/administrative variables and directly measured anthropometric variables. These variables consistent of both continuous and categorical measures, with the categorical measures initially stored as string variables. Demographic/administrative variables include personal demographic characteristics, army-related information of the subjects (such as branch), self-reported height and weight, and information about where and when the measurements were taken.[1] The anthropometric variables include measures related to stature and vertical height, limb length and reach, breadth and circumference, and measured weight and height.[1,2]

The following sections of this report contain the overall analysis plan, data quality assessment, and exploratory analysis findings. A copy of the project's .do file with all project code and annotations can be found in the appendix.

2. Data Preparation

Step 1: Set up project structure

- Create Project 1 folder
- Save the following in the Project 1 folder:
    - Original dataset
    - Raw backup copy
    - Project instructions
    - Project report document
    - ANSUR2 MFR and TR11 pdfs
    - Do file
    - Log file:
- Create and annotate do file

Step 2: Import and inspect dataset + rename variables if need) + add labels to variables

- Open dataset; save working version; turn on log
- Look at variable names and labels and see if any need to be renamed to understand what they are or if category labels are needed. Document in do file. Refer to the MFR and TR11 pdfs as needed
- Add labels to variables, including the values for categorical variables
- Get oriented to the dataset; use describe, codebook, summarize, tab, browse, etc.
- Note findings

Step 3: Create a unique key for each observation

- Generate a unique ID and verify uniqueness
- Save dataset

Step 4: Recode missing values and change measures to cm or kg

- Recode the following numbers to Stata special missing values per assignment instructions:
    - -77 = .a
    - -88 = .b
    - -99 = .c
- Change anthropometric measures from inches/mm to cm and lbs to kg; update labels
- Check work, document in do file
- Save dataset

Step 5: Inspect for unreasonable or suspect values and address them with flag variables and special missing values

- Look at each variable using sum and tab
- Look for biologically implausible values, negative values, extremely low or high values. Recode to special missing or create a flag based on a set criteria.
    - Document reasons for flag and criteria used in do file and verification of checking
    - Recode values to Stata special missing values and document reasons for recodes, criteria used, and what any new special missing value letters mean
- Add labels if needed
- Drop underage subjects
- Check work, document in do file
- Save dataset

Step 6: Remove underage subjects

- Look at age distribution
- Identify any observations under age 18 and create a flag variable to document underage subjects
- Note number of underage in do file
- Drop underage subjects

- Document in do file
- Save dataset

Step 7: Check and resolve duplicates

- Look for duplicates and create new flag variable for ones I don't want to exclude but are suspicious. Look for duplicates based on matching combinations of anthropometric measures and demographics.
- Address subject id duplicates to be excluded and document decision rules for which measures will be kept, recode issues using special missing values.
- Create dataset for duplicates and pull the duplicates out into the separate dataset.
- Remove duplicates from main working dataset
- Check work and document in do file
- Save dataset

Step 8: Create BMI variables

- Look up BMI formula
- Create new variables for height and weight that have the appropriate units. Make sure to protect missing values. Document in do file.
- Create the continuous bmi variable using gen and replace and the bmi formula. Make sure to protect missing values. Have it be numerical and add labels. Check things. Document in do file.
- Label bmi variable
- Check bmi variable for any implausible values. Document in do file.
- Create bmi_cat variable using gen and replace. Make sure to protect missing values. Check things. Document in do file
- Label bmi_cat and levels of bmi_cat
- Check things. Document in do file.
- Save dataset

Step 9: Create season variable

- Create a categorical variable for the season in which the taken, based on the date variable
- Inspect the data variable to see how its stored/the format
- Make a new variable that has date in a Stata data format. Make sure to protect missing values
- Extract the month of the date and create a month variable
- Create a season variable using standard season definitions and document in do file. Make sure to protect missing values
    - Use numbers 1-4 to code the seasons
    - Label season variable
    - Label levels of the season variable with fall, winter, spring summer
- Check work and document in do file
- Save dataset

Step 10: Create gender categorical variable

- Look at gender to look at storage type. Use encode to create a new numeric categorical variable. Make sure to protect missing values
- Name the new variable gender_cat
- Label categorical levels
- Check work and document in do file
- Save dataset

Step 11: Create preferred hand categorical variable

- Look at storage type. Use encode to create a new numeric categorical variable. Make sure to protect missing values
- Name the new variable pref_hand_cat
- Label categorical levels
- Check work and document in do file
- Save dataset

Step 12: Create body type categories variable

- Determine body type categories for males and females based on height, weight, and at least two other variables.
    - Create a vertical proportion and build index that based on ratios
        - Assign vertical proportion and build categories by sex specific tertiles
    - Create body categories based on combination of vertical proportion and build categories
- Check work and document in do file
- Save dataset

Step 13: Create t shirt size variable

- Find a unisex t shirt sizing chart based on chest size
- Create t shirt size numeric variable with numbers 1-6 using t shirt sizing chart as the cut points. Protect missing values.
- Check work to make sure size aligns with chest inches ranges
- Add labels for variable and variable levels
- Check work
- Save dataset

Step 14: Create anthropometric measures table

- For each anthropometric variable, determine mean, SD, min, and max. Divide out by sex.
- Determine number of missing/suspicious values for each variable
- Create table in Excel from the Stata outputs
    - Organize by body part
- Copy table into assignment report and format

Step 15: Create figure height attributable to height to hip by sex

- Identify what variables needed
- Create variable pct_hip_height variable, label, check distribution and address any suspicious values
- Save dataset
- Create a bar graph focused on mean with error bars comparing males and females
- Create box plot to look at distributions
- Format figures
- Copy figure to report
- Write paragraph

Step 16: Explore correlation of stature variables and create scatterplot

- Identify stature related measures from correlation table
    - Use Pearson correlation to identify the 2 measures that have the highest linear correlation - look at correlation coefficients and significance levels
- Create a scatterplot with the 2 measures
- Format scatterplot
- Address overplotting if necessary
- Copy figure to report and format
- Write paragraph

Step 17: Explore correlation of stature variables by sex and create table

- Identify stature related measures from correlation table
    - Use Pearson correlation to identify the 2 measures that have the highest linear correlation - look at correlation coefficients and significance levels; use pwcorr and stratify by sex
- Extract correlations between stature and each variable for each sex
- Create table in word that shows r values by sex
- Interpret magnitude of r and compare patterns by sex
- Copy table to report
- Write paragraph

Step 18: Create variables with comparable units and difference variables

- Create a weight difference variable with reported weight - measured weight.
    - Label variable.
    - If it's negative, they underreported.
- Create a height difference variable with reported height - measured height.
    - Label variable.
    - If it's negative, they underreported.

Step 19: Examine distribution of differences for actual and reported weight

- Look at mean, median, SD, quartiles, range for weight difference variable. Use summarize.

- Create a box plot using the weight difference variable above
- Create a hexplot to explore deeper.
- Copy to report and format.
- Create bar graph using the weight difference variable above to show average differences by gender.
    - Consider running a t-test or a regression to get an odds ratio.
- Copy to report and format.

Step 20: Examine distribution of differences for actual and reported height

- Look at mean, median, SD, quartiles, range for height difference variable. Use summarize.
- Create a boxplot using the height difference variable above
- Format boxplot
- Create a hexplot to explore deeper.
- Copy to report and format
- Write summary of findings for section 4.2

Step 21: Examine distribution of body types and weight and BMI

- Create violin plots for body types across weight and BMI.
- Create a table to show the frequency of body types for each BMI category.
    - Use cross tabs
    - Export to Excel and format
- Write paragraph

Step 22: Create table for association of BMI and age (continuous variables), stratified by sex

- Create scatterplots of BMI and age, with one for each sex
- Calculate correlations using a command for each sex category and add to figures.
- Run linear regression and create table with output in Excel.
- Copy table and scatterplots to report and format.
- Write paragraph.

Step 23: Create table for association of BMI and age (categorical variables), stratified by sex

- Create age category variable using NCHS standard age categories. Have it be numerical with labels.
    - Protect missing values
    - Add labels
    - Check work
    - Document cut points and checking in do file
- Save dataset
- Table creation:
    - Use cross tabulations to look at counts/frequencies in each category

    - Run a chi square to get percentage in each for BMI and age categorical variables
    - Create table in Excel from output
- Figure creation:
    - Create stacked bar charts (one for males and one for females) showing BMI weight categories as a percentage across age groups
    - Format
- Copy table and figures to report and format
- Write paragraph

Step 24: Compare 5.a vs. 5.b

- Write comparison paragraph of the two analyses

3. <u>Sample Characteristics</u>

Anthropometric Characteristics

Table 3.1. Anthropometric characteristics of ANSUR II Subjects

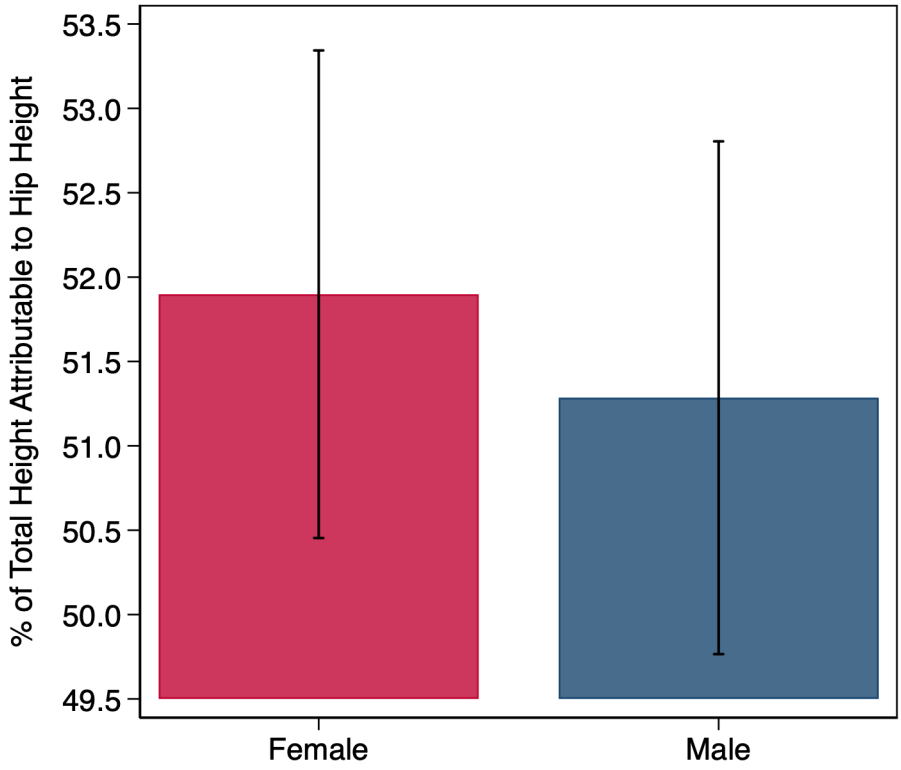
	Mean	SD	Min	Max
Overall Body Size				
Measured Stature (cm)	171.441	9.002	140.900	199.300
Self-Reported Stature (cm)	173.375	9.810	142.240	238.760
Measured Weight (kg)	79.707	15.654	35.800	144.200
Self-Reported Weight (kg)	79.266	15.246	39.002	145.578
Upper Body				
Cervicale Height (cm)	147.750	8.432	118.400	173.800
Waist Height (Omphalion) (cm)	103.150	6.270	80.500	124.500
Arm Span (cm)	176.380	11.084	132.300	212.100
Thumbtip Reach (cm)	78.970	5.382	60.200	99.800
Chest Circumference (cm)	102.210	10.067	69.500	146.900
Waist Circumference (cm)	91.452	11.427	61.100	137.900
Lower Body				
Trochanterion Height (cm)	88.274	5.440	66.000	109.500
Knee Height (Midpatella) (cm)	47.551	3.308	34.600	61.200
Functional Leg Length (cm)	110.205	6.855	83.500	131.600
Bicristal Breadth (cm)	27.474	1.922	19.700	36.200
Hip Breadth (cm)	34.838	2.530	26.400	47.300
Hip Breadth (Sitting) (cm)	38.573	3.240	28.000	54.100
Foot Length (cm)	26.303	1.739	19.800	32.300

Table 3.2. Summary dataset values of anthropometric measures collected in the ANSUR II Survey

	N Observed	N Missing	N Suspicious[a]
Overall Body Size			
Measured Stature (cm)	6,068	0	0
Self-Reported Stature (cm)	6,068	0	14
Measured Weight (kg)	6,068	0	0
Self-Reported Weight (kg)	6,067	1	107
Upper Body			
Cervicale Height (cm)	6,068	0	0
Waist Height (Omphalion) (cm)	6,068	0	0
Arm Span (cm)	6,068	0	0
Thumbtip Reach (cm)	6,068	0	0
Chest Circumference (cm)	6,068	0	0
Waist Circumference (cm)	6,068	0	125
Lower Body			
Trochanterion Height (cm)	6,068	0	0
Knee Height (Midpatella) (cm)	6,068	0	0
Functional Leg Length (cm)	6,068	0	0
Bicristal Breadth (cm)	6,068	0	0
Hip Breadth (cm)	6,068	0	0
Hip Breadth (Sitting) (cm)	6,068	0	0
Foot Length (cm)	6,068	0	36

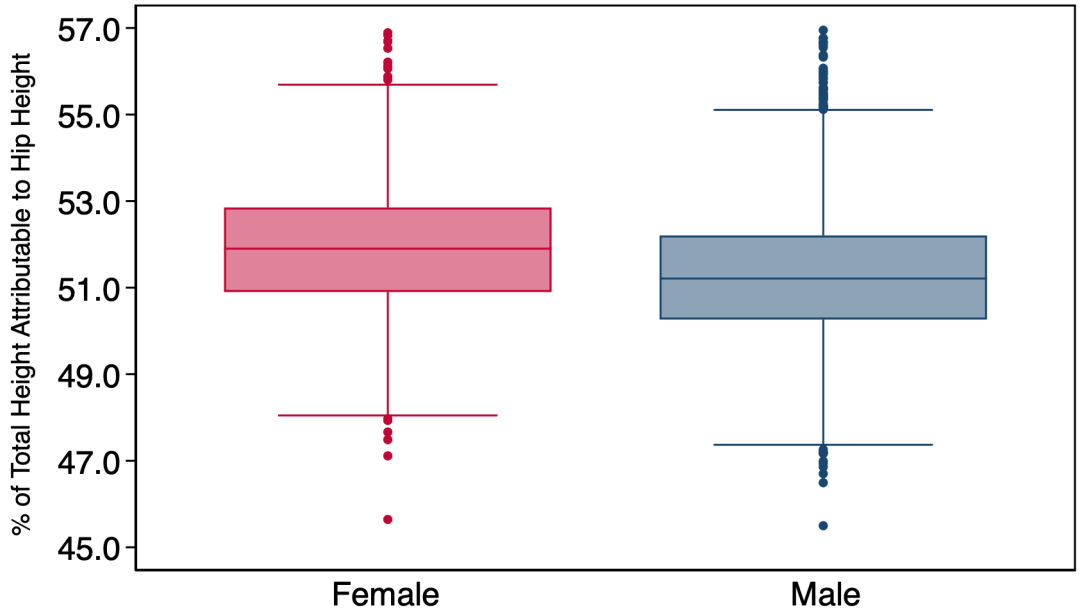
[a] See the Data Quality Assessment section below for information on how suspicious values were identified

Figure 3.1 Mean percent of total height attributable to hip height by gender



Data were collected from ANSUR II survey subjects. Bars (pink = female, blue = male) represent the mean percentage of total height attributable to hip height measured at the trochanterion. Error bars show ±1 standard deviation. % = percent of total height

Figure 3.2. Distribution of percent of total height attributable to hip height by gender



Data were collected from ANSUR II survey subjects. The % of total height to hip refers percentage of total height attributable to hip height measured at the trochanterion. Boxes (pink = female, blue = male) represent the interquartile range (IQR) with the median shown as the center line. Whiskers extend to the minimum and maximum values within 1.5 × IQR. Points beyond the whiskers represent values outside 1.5 x the IQR.

Figure 3.1 and Figure 3.2 Summary

Figures 3.1 and 3.2 show that hip height (measured at the trochanterion) accounted for just over half of total height in both male and female ANUR II subjects. Females had a mean percent of total height attributable to hip height of 51.9%, and males had a mean of 51.3%. Although females had a slightly higher percent of total height attributable to hip height (0.6% mean difference), the standard deviations for both genders (female = 1.4%; male =1.5%) were more than twice the magnitude of the mean difference. This suggests the variability within genders is more than the variability between genders. The relatively similar standard deviations for each gender also indicate that individual subjects within each gender typically vary by a similar amount around their respective means (though males have slightly more variability). The boxplots (Figure 3.1) show substantial overlap in the interquartile ranges, which further supports similarity in in the distributions.

Data Quality Assessment

Data quality was assessed and addressed by examining completeness, consistency, accuracy, and validity through the review of data distributions, identification of missing values, assessment of string value consistency, verification of categorical coding, and evaluation of duplicate records. The dataset was determined to be nearly complete, internally consistent, and contain only biologically plausible anthropometric values. All recodes, generated flags, and duplicate resolutions were verified for accuracy, and all changes were documented within the project .do file (see appendix).

The dataset initially lacked variable and value labels, which were added to be consistent with official ANSUR II documentation.[1,2] Categorical variables were reviewed for consistency with the official ANSUR II documentation and for text differences in string values. The subject numeric race and Department of Defense race variables contained invalid values per official documentation and were recoded to special missing value .g (see Table 3.3).[1] The writing preference string variable was also standardized to ensure consistent spelling and formatting. All anthropometric measures were converted to the appropriate metric unit. Subjects under age 18 were removed from the dataset because they did not meet age eligibility criteria.[1]

Various values were recoded as special missing values to indicate 1) missing values in the initial dataset (recoded to .a, .b., or .c), 2) values that were determined to be biologically implausible or sufficiently suspicious to warrant recoding (recoded to .d or .f), or 3) missing values in a derived formula-based variable (recoded to .h or .i) using a value that had been previously recoded to .d or .f (see Table 3.3). Recoding cut point thresholds for thump tip rich and measured weight were determined based on logical reasoning of biological limits and the data distribution (see Table 3.3). Suspicious but biologically plausible values were flagged but retained in the dataset (see Table 3.4). Flag cut point thresholds for such values were based on Army enlistment requirements for height and weight, foot length measures based on common retail-shoe sizing, dataset distribution-based waist circumference extremes, and standard BMI classifications (see Table 3.4).[3-5] Blank string values for date and ethnicity were also identified and flagged (see Table 3.4).

No exact duplicates across all variables were identified. However, many observations were exact matches based on component, installation, branch, stature, measured weight, waist circumference, hip breadth, and hip breadth sitting, which would be extremely unlikely for different subjects. Inspection of these duplicate pairs also showed one observation to have a missing date. When date was included in the duplicate report criteria, the number of exact matches dropped to zero, suggesting a date-related duplication issue. Additionally, many observations with a measured weight that had been recoded to .f (suspiciously large) were observed to have a missing date and identical anthropometric measures, suggesting a weight-related duplication issue. One observation (ID 6515) was also missing measured anthropometrics but matched another subject (ID 825) on all demographic, army-related, and self-reported variables, suggesting an incomplete assessment-related duplication issue. Duplicates were resolved by retaining the duplicate observation with a valid date, non-suspiciously large weight value, or completed anthropometric measures. After resolving all

duplicates, the dataset contained 6,068 observations which is consistent with official ANSUR II documentation.[1] Nearly all missing values were also resolved (see Table 3.2).

Table 3.3. Special missing values and recoding decisions

Special Missing Value	Original Value	Meaning	Variables Affected	Rationale / Cut Points
.a	-77	Not recorded	None	Per assignment instructions
.b	-88	Refused measurement	None	Per assignment instructions
.c	-99	Unknown	• Thump tip reach • Span • Knee height midpatella • Waist heigh ompphalion • Functional leg length • Trochanterion height • Stature • Waist circumference • Chest circumference • Bicristal breadth • Hip breadth • Hip breadth sitting • Weight	Per assignment instructions
.d	0	Biologically implausible	Self-reported weight	Zero is outside of biological limits
.e	Suspiciously small non-zero value (see rationale column)	Suspiciously small	None	N/A , no variables impacted were identified
.f	Suspiciously large value (see rationale column)	Suspiciously large	• Thumbtip reach • Measured weight	Above realistic/plausible upper bound • Weight > 400 kg • Thumbtip reach > 150 cm
.g	Invalid categorical code	Invalid code	• Subject numeric race • Department of Defense race	Per survey memorandum for record[1] • Subject numeric race > 8 • Department of Defense race ≠ 1-6, 8
.h	. (for date variable only)	Structurally missing date	Date	Distinguished from generic "." for duplicate resolution
.i	. (for BMI variables only)	Missing measured stature or weight	• BMI (numerical) • BMI (category)	BMI cannot be calculated without a stature or weight measurement

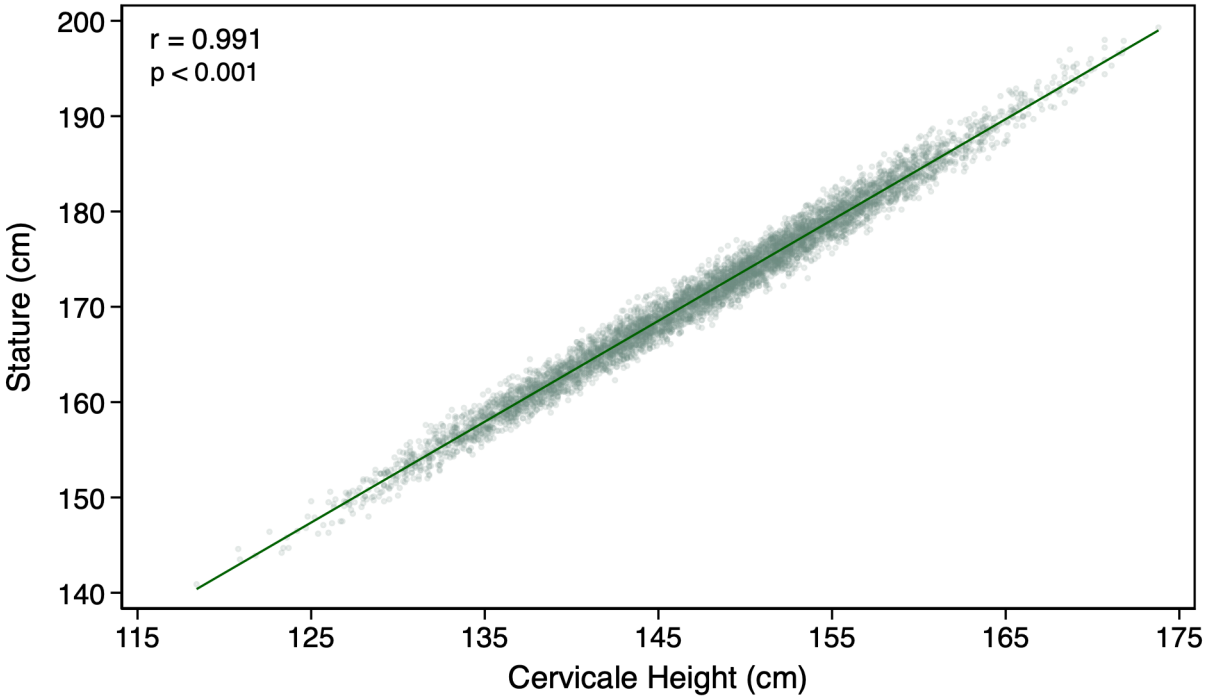
Table 3.4. Flag variables and decisions

Flag Variable	Cut Points	Variables Affected	Cut Point Rationale	Action Taken
flag_weightkg_selfreport	Male and Female < 41.28 kg Male > 113.398 kg Female > 107.048 kg	Weight (self-reported)	Army enlistment requirements for minimum weight and maximimum weight for tallest/oldest subjects[3]	Flagged only
flag_heightcm (self reported height)	< 147.32 cm > 203.2 cm	Height (self Reported)	Army enlistment requirements for minimum and maximum height[3]	Flagged only
flag_stature (measured height)	< 147.32 cm > 203.2 cm	Stature	Army enlistment requirements for minimum and maximum height[3]	Flagged only
flag_footlength	Male < 23 cm or > 31.5 cm Female < 22 cm or > 27.9 cm	Foot length	Retail shoe size standards (HOKA)[4]	Flagged only
flag_waistcircumference	< 1st percentile or > 99th percentile Male < 72.5 cm or > 121.4 cm Female < 66.5 cm or > 111.3 cm	Waist Circumference	Distribution-based threshold	Flagged only
flag_bmi	BMI value < 18.5 or > 35	BMI (continuous)	CDC underweight / obese II threshold[5]	Flagged only
flag_bmi_cat	Underweight or Obese II+ category (BMI value < 18.5 or > 35)	BMI (categorical)	CDC underweight / obese II threshold[5]	Flagged only
flag_ethnicity_missing	Blank string ethnicity	Ethnicity	Missing in dataset	Flagged only
flag_strdate_missing	Blank string date	Date (string)	Missing in dataset	Flagged only
flag_duplicates1	Duplicates based on: component, gender, age, stature, weightkg, waist circumference, hip breadth, hip breadth sitting, installation, branch	N/A	Identical measures across records	Duplicates dropped
flag_duplicates_weighterror	Duplicates based on: component, gender, age, installation, branch, thumb tip reach, span, foot length, waist height (omphalion), functional leg length, cervicale height, trochanterion height, stature, waist circumference, chest circumference, bicristal breadth, hip breadth, hip breadth sitting	N/A	Identical antropometric measures	Duplicates dropped
flag_duplicates_incomplete	ID = 6515 or 825	N/A	Identified 6515 as incomplete duplicate match to 825	Duplicate dropped

4. Relationships Between Anthropometric Measures

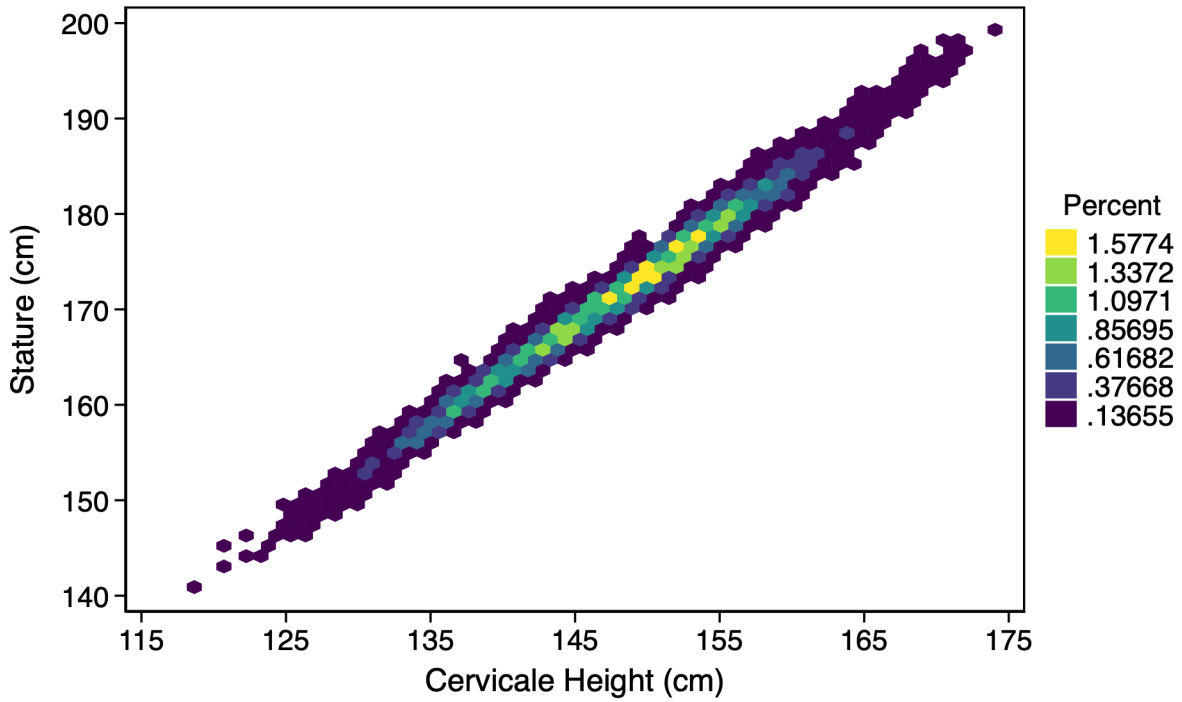
Relationship Between Measures of Stature

Figure 4.1. Relationship between stature and cervicale height



Data were collected from ANSUR II subjects. The scatterplot displays measured stature (cm) plotted against cervicale height (cm). Each teal marker represents an individual observation. The solid green line shows the ordinary least squares best-fit line. The Pearson correlation coefficient (r) and the p-value are shown on the figure. cm = centimeters

Figure 4.2. Density relationship between stature and cervicale height



Data were collected from ANSUR II subjects. The plot displays measured stature (cm) plotted against cervicale height (cm). Each hexagon shows the percent of total observations at a given stature and cervicale height range, with purple indicating the lowest percentage and yellow indicating the highest. cm = centimeters

Figure 4.1 and Figure 4.2 Summary

Among the measures of stature (stature, knee height midpatella, cervicale height, trochanterion height, waist height omphalion, functional leg length, foot length, and thumb tip reach), the two measures that were most highly correlated were stature and cervicale height. The scatterplot in Figure 4.1 shows that stature increases as cervicale height increases, and observations are clustered near the ordinary least squares best-fit line. The Pearson correlation coefficient ($r = 0.991$, $p < 0.001$) indicates a nearly perfect and statistically significant linear association between the measures.

Due to the large sample size and clustering, overplotting is present in the scatterplot. To provide an additional visualization of the relationship between stature and cervicale height, a density hexplot was also created (Figure 4.2). The clustering of lighter colors in the center of the diagonal band supports the strength and linearity of the association but also shows that stature and cervicale height measures are clustered around mid-range values.

Table 4.1. Correlation of stature and other measure of stature by sex

	Pearson Correlation Coefficient (r)			
Measure of Stature	**Sample**	**Female**	**Male**	**Difference[a]**
Cervicale Height (cm)	0.991	0.984	0.984	0.000
Waist Height (Omphalion) (cm)	0.937	0.909	0.909	0.000
Span (cm)	0.899	0.819	0.823	0.004
Knee Height (Midpatella) (cm)	0.889	0.843	0.833	-0.010
Functional Leg Length (cm)	0.888	0.832	0.818	-0.014
Trochanterion Height (cm)	0.877	0.858	0.850	-0.008
Foot Length (cm)	0.845	0.722	0.718	-0.004
Thumbtip Reach (cm)	0.806	0.671	0.691	0.020

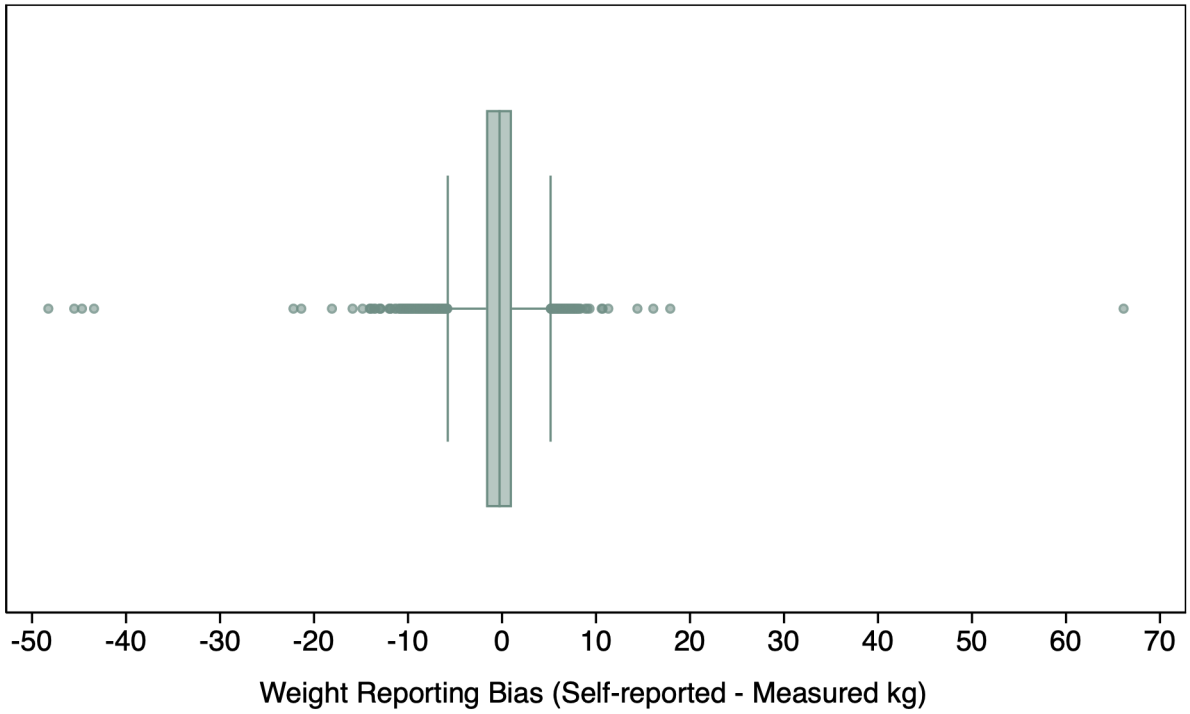
[a] Difference = Male - Female

Table 4.1 Summary

The correlation of stature and other measures of stature is shown above in Table 4.1, with measures listed in the order of greatest correlation for the full sample, followed by female and male subjects. Cervicale height demonstrated the strongest association with stature in both females and males, followed by waist height (omphalaion). The correlation coefficients for these measures with stature were nearly identical for both genders, differing only at the ten thousandths place. For knee height (midpatella), functional leg length, trochanterion height, and foot length, males had slightly lower correlations with stature compared to females (range of r difference: 0.008 - 0.014). For span and thumbtip reach, females had slightly lower correlations with stature compared to males (range of r difference: 0.004 - 0.020). Overall, both the magnitude and direction of the correlations were highly similar between females and males, indicating the linear relationship between stature and other measures of stature were consistent across both sexes.

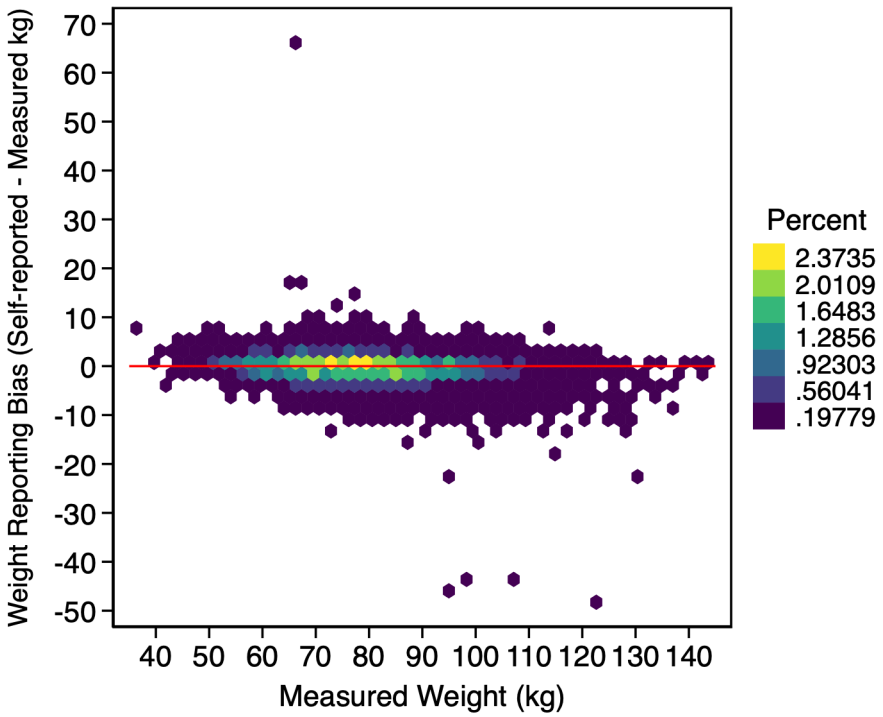
Relationship Between Actual and Reported Height and Weight

Figure 4.3. Distribution of weight reporting bias



Data were collected from ANSUR II survey subjects. Weight reporting bias refers the difference between the weight the subject reported and their measured weight in kg. The box represents the interquartile range (IQR) with the median shown as the center line. Whiskers extend to the minimum and maximum values within 1.5 × IQR. Points beyond the whiskers represent values outside 1.5 x the IQR. kg = kilograms

Figure 4.4. Density relationship between weight reporting bias and measured weight

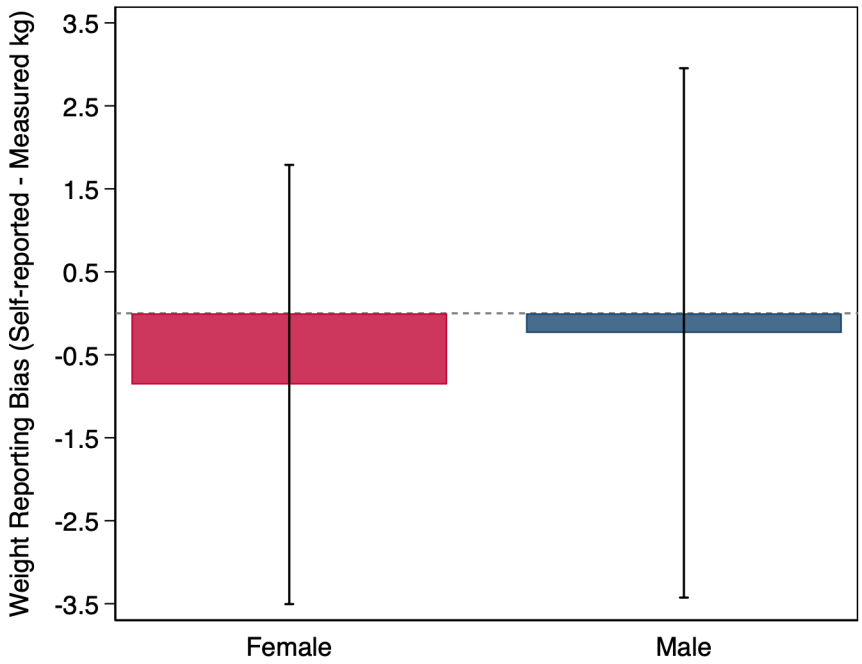


Data were collected from ANSUR II subjects. The plot displays weight reporting bias (difference between the weight the subject reported and their measured weight in kg) plotted against their measured weight. Each hexagon shows the percent of total observations at a weight reporting bias range and measured weight range, with purple indicating the lowest percentage and yellow indicating the highest. The red line at zero is the reference line for reported weight matching measured weight. kg = kilograms

Figure 4.3 and Figure 4.4 Summary

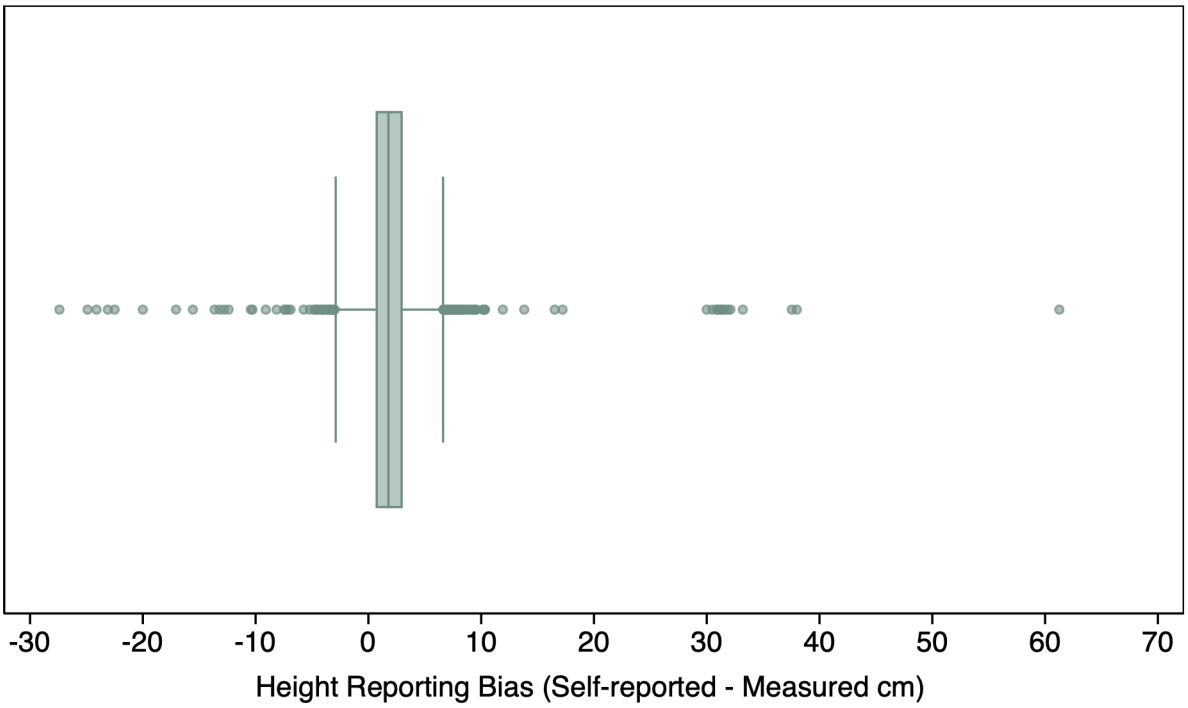
Figures 4.3 and 4.4 show the distribution of weight reporting bias, defined by reported weight minus measured weight. The mean weight reporting bias was -0.44 (SD = 3.04 kg), indicating that on average, subjects underreported weight. The interquartile range extended from -1.69 kg to 1.05 kg, indicating that 50% of subjects reported their weight within this range of proximity to their measured weight. The distribution was left skewed (skewness = -0.89), which suggests a greater frequency of underreporting compared to overreporting. Although most weight reporting bias values were relatively close to zero, there are extreme values in both over and underreporting. Figure 4.4 provides insights into the distribution of weight reporting bias by weight and reveals that subjects at the low end of weight ranges do not tend to under report weight, while subjects at the high end of weight ranges do not tend to over report weight.

Figure 4.5. Weight reporting bias by sex



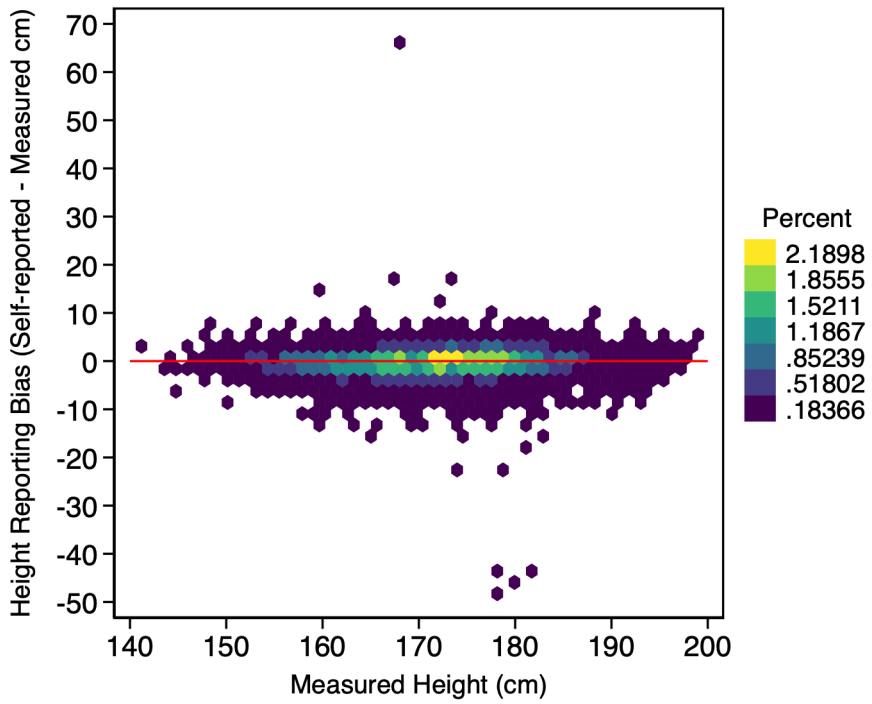
Data were collected from ANSUR II survey subjects. Bars (pink = female, blue = male) represent the mean weight reporting bias (difference between the weight the subject reported and their measured weight in kg). Error bars show ±1 standard deviation. The dashed gray line at zero is the reference line for reported weight matching measured weight. kg = kilograms

Figure 4.6. Distribution of height reporting bias



Data were collected from ANSUR II survey subjects. Height reporting bias refers the difference between the height the subject reported and their measured height in cm. The box represents the interquartile range (IQR) with the median shown as the center line. Whiskers extend to the minimum and maximum values within 1.5 × IQR. Points beyond the whiskers represent values outside 1.5 x the IQR. cm = centimeters

Figure 4.7. Density relationship between height reporting bias and measured height



Data were collected from ANSUR II subjects. The plot displays height reporting bias (difference between the height the subject reported and their measured height in cm) plotted against their measured height. Each hexagon shows the percent of total observations at a height reporting bias range and measured height range, with purple indicating the lowest percentage and yellow indicating the highest. The red line at zero is the reference line for reported height matching measured height. cm = centimeters

Figure 4.6 and Figure 4.7 Summary

Figures 4.6 and 4.7 show the distribution of height reporting bias, defined by self-reported height minus measured height. The mean height reporting bias was 1.93 cm (SD = 2.74 cm), indicating that on average, subjects overreported their height. The interquartile range extended from 0.64 cm to 3.04 cm, indicating that 50% of subjects reported their height within this proximity to their measured height. The distribution was strongly right skewed (skewness = 3.94), which suggests a greater frequency of overreporting compared to underreporting. Although most height reporting bias values were relatively close to zero, there are extreme values in both over and underreporting. Figure 4.7 provides insights into the distribution of height reporting bias, showing that reporting bias remains relatively consistent across different heights.

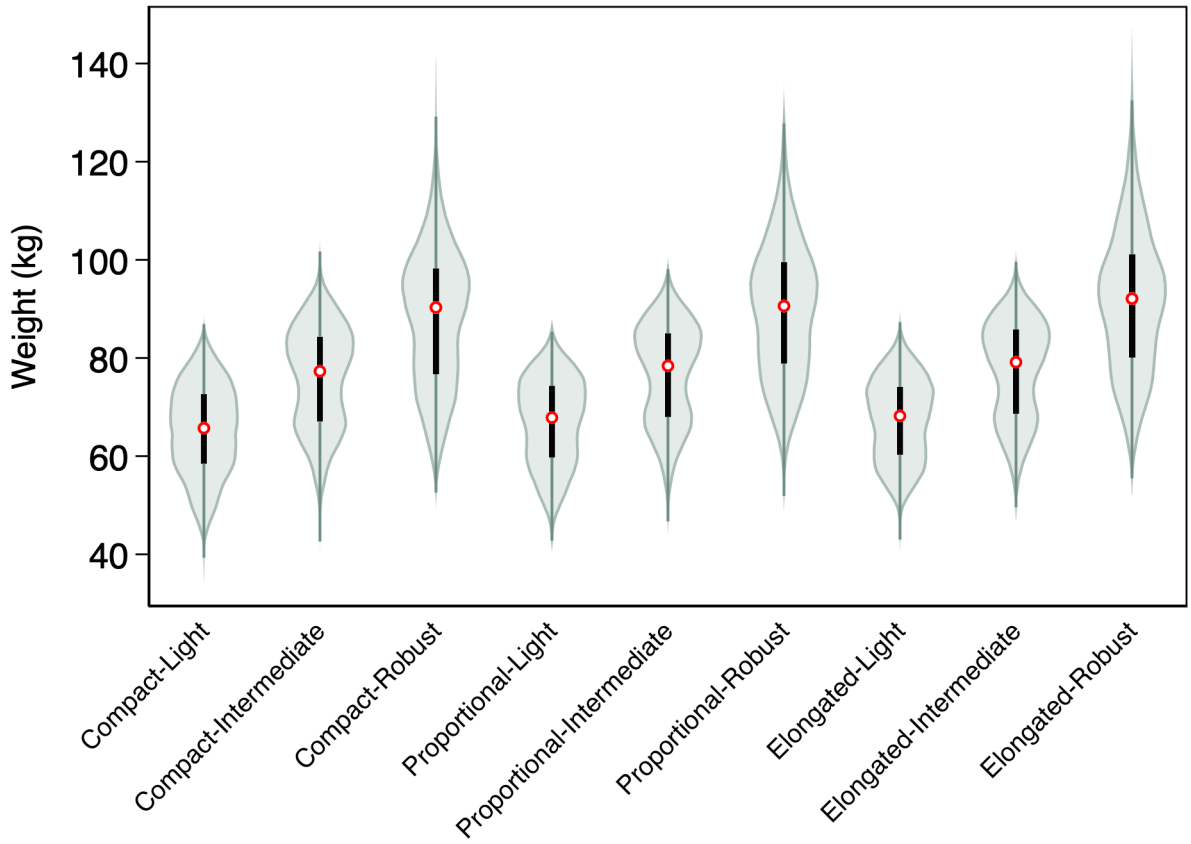
Summary of Weight and Height Reporting Biases

Taken together, the self-reported and measured weight and height comparisons in Figures 4.3-4.7 suggest there would be directional bias in using reported weight and height in lieu of measuring weight and height. Weight tends to be underreported, while height tends to be overreported. The sex-specific analysis in Figure 4.4 also shows that females underreported

their weight by a greater amount on average (-0.86 kg) than males (-0.24 kg). Further analyses showed that males had significantly lower odds of underreporting weight compared to females (OR = 0.589, 95% CI 0.528 – 0.658, $p < 0.001$). Figures 4.3 and 4.6 also suggest that self-reported weight bias may vary across different measured weights, while self-reported height-biases appears to be relatively stable across measured heights. Overall, these findings suggest that using reported weight and height in lieu of measured weight and height could be problematic if weight and height measures need to have a height degree of accuracy for either military activities or data analyses.

Relationship Between Derived 'Body Type' and Weight/BMI

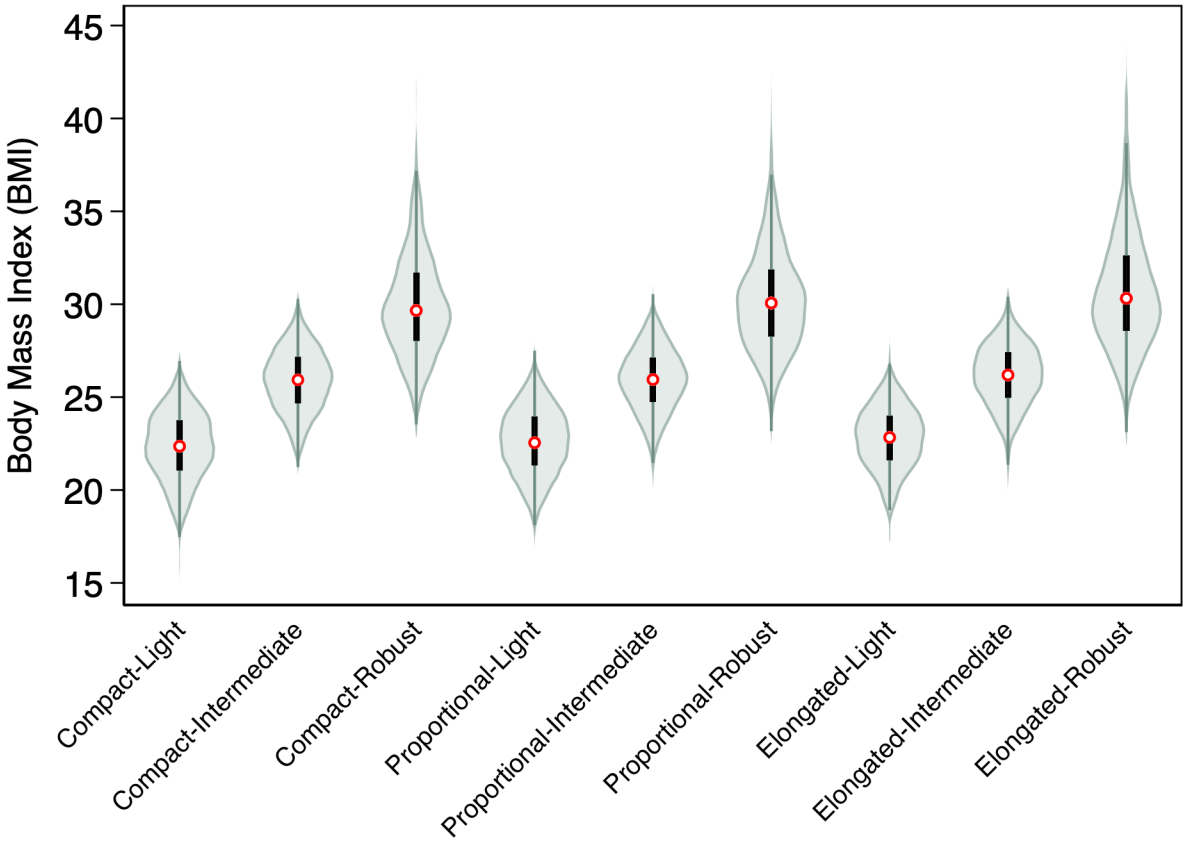
Figure 4.8. Distribution of weight by body type



Data were collected from ANSUR II subjects. The violin plots display the distribution of measured weight (kg) across nine body type categories. The width of each violin shows the kernel density of the data at a given weight value, with wider sections indicating a higher concentration of observations. The black lines represent the interquartile range (IQR), with the median shown as the red circle and the teal center line extending to the minimum and maximum values within 1.5 × IQR. Body type categories were defined based on nine combinations of vertical proportion and build. Vertical proportion was determined using sex-specific tertiles of leg-to-stature ratio (LSR). Compact included the lowest tertile, proportional included the middle tertile, and elongated included the highest tertile. Build was determined by using sex-specific tertiles of mass-to-height ratio (HMR) and

trunk-to-stature ratio (HSR). Trunk was defined by waist circumference. Light builds included the lowest tertile of both HMR and TSR, robust builds included the highest tertile of HMR or TSR, and intermediate builds included all other HMR and TSR combinations. kg = kilograms

Figure 4.9. Distribution of body mass index (BMI) by body type



Data were collected from ANSUR II subjects. The violin plots display the distribution of body mass index (BMI) across nine body type categories. The width of each violin shows the kernel density of the data at a given BMI value, with wider sections indicating a higher concentration of observations. The black lines represent the interquartile range (IQR), with the median shown as the red circle and the teal center line extending to the minimum and maximum values within 1.5 × IQR. Body type categories were defined based on nine combinations of vertical proportion and build. Vertical proportion was determined using sex-specific tertiles of leg-to-stature ratio (LSR). Compact included the lowest tertile, proportional included the middle tertile, and elongated included the highest tertile. Build was determined by using sex-specific tertiles of mass-to-height ratio (HMR) and trunk-to-stature ratio (HSR). Trunk was defined by waist circumference. Light builds included the lowest tertile of both HMR and TSR, robust builds included the highest tertile of HMR or TSR, and intermediate builds included all other HMR and TSR combinations.

Table 4.2. Percentage of body types in each body mass index category

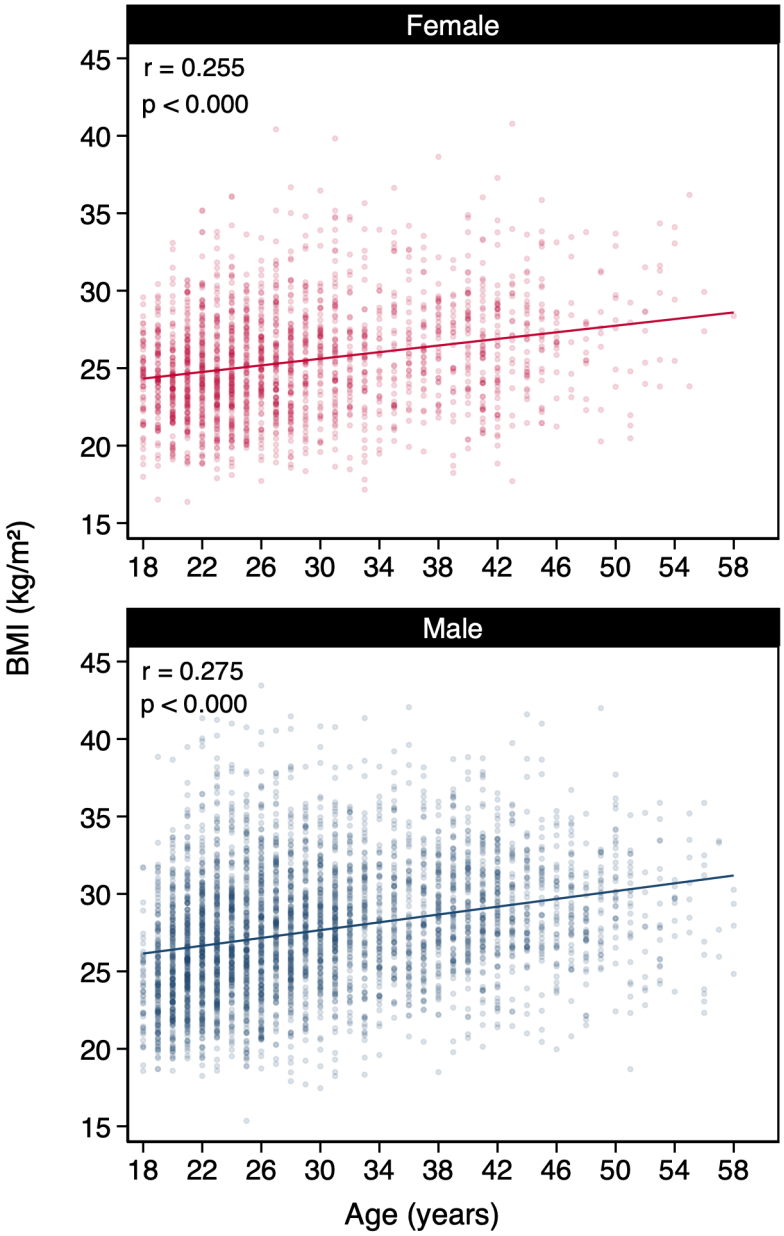
	Body Mass Index (BMI) Category			
Body Type	**Underweight (%)**	**Healthy Weight (%)**	**Overweight (%)**	**Obese (%)**
Compact				
Light	2.09	91.20	6.70	0.00
Intermediate	0.00	30.79	68.79	0.42
Robust	0.00	1.71	53.92	44.37
Proportional				
Light	1.02	90.24	8.74	0.00
Intermediate	0.00	29.72	70.00	0.28
Robust	0.00	1.60	47.84	50.55
Elongated				
Light	0.65	90.29	9.06	0.00
Intermediate	0.00	25.53	73.59	0.88
Robust	0.00	1.57	44.45	53.97

Figure 4.8, Figure 4.9, and Table 4.2 Summary

Figure 4.8, Figure 4.9, and Table 4.2 demonstrate that the body type classification used in this exploratory data analysis differentiates differences in both weight and obesity relatively well. In respect to weight classifications by BMI, overweight subjects most commonly fell into the intermediate build, and obese subjects most commonly fell into a robust build across all vertical proportion categories (compact, proportional, and elongated) (Table 4.2). No underweight subjects were classified as intermediate or robust builds across the vertical proportion categories, indicating a strong separation of categories for lower and higher BMIs. The violin plots in Figures 4.8 and 4.9 support this observation. Within each vertical proportion category, the median weight and BMI increase in a stepwise fashion from light to intermediate to robust builds. The weight distributions show a greater spread and overlap across body type groups (Figure 4.8), whereas the BMI distributions are more normally shaped and have narrower interquartile ranges and less interquartile range overlap within each vertical proportion category (Figure 4.9). Additionally, when comparing BMI and weight across vertical proportions within the same build (e.g., compact-light vs. elongated-light), the distributions are more similar, suggesting that vertical proportion contributes less to distinguishing weight and obesity differences than build within the body type classification system used. Overall, the body type classification performs relatively well in identifying overweight and obese subjects.

5. Relationships Between Demographic Characteristics and BMI

Figure 5.1. Relationship between age and body mass index by sex



Data were collected from ANSUR II subjects. The scatterplots display body mass index (BMI) plotted against age in years for each sex. Each pink (female) and blue (male) marker represent an individual observation. The solid line shows the ordinary least squares best-fit line. The Pearson correlation coefficient (r) and the p-value are shown on the figure. kg/m^2 = kilograms per meter squared

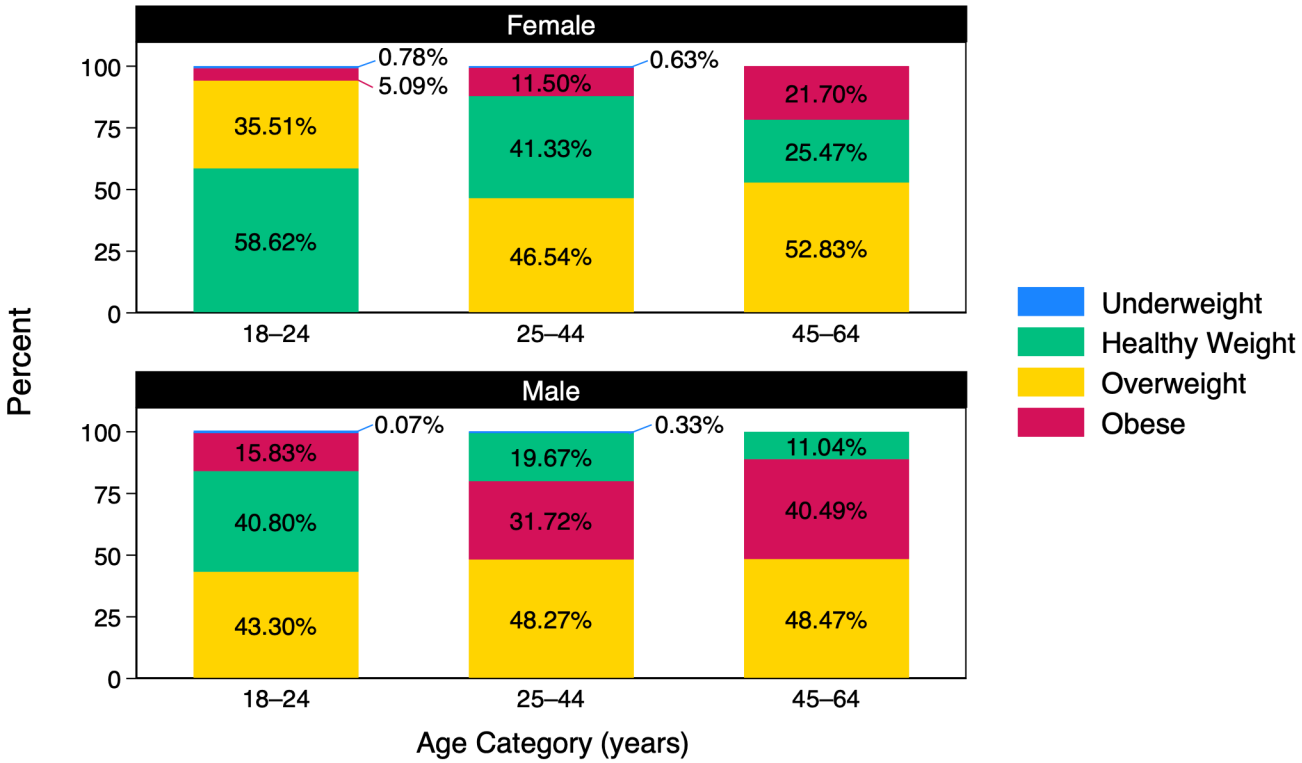
Table 5.1. Association between age and body mass index by sex

Sex	N	B Coefficient	Standard Error	95% CI	p-value	R^2
Female	1,985	0.107	0.009	0.089 – 0.125	0.000	0.065
Male	4,083	0.126	0.007	0.112 – 0.140	0.000	0.076

Figure 5.1 and Table 5.1 Summary

Figure 5.1 shows a weak positive association between age and BMI for both males ($r = 0.275$, $p < 0.001$) and females ($r = 0.255$, $p < 0.001$). The scatterplots show a gradual upward trend in BMI as age increases and have a wide dispersion of points around the ordinary least squares best fit line for both sexes, indicating substantial variability in BMI for a given age. Among females, BMI increased by 0.107 kg/m^2 per year increase in age (95% CI: 0.089-0.125, $p < 0.001$), while BMI increased slightly more at 0.126 kg/m^2 per year (95% CI: 0.112-0.140, $p < 0.001$) (see Table 5.1). Although the associations are statistically significant, the magnitude of R^2 suggests that age explains a small proportion of BMI variability. Overall, the relationship between age and BMI is similar across sexes, with a slightly stronger association observed among males.

Figure 5.2. Distribution of body mass index categories across age groups by sex



Data were collected from ANSUR II subjects. Stacked bar charts show the percentage of subjects across weight status categories defined by standard body mass index (BMI) category ranges, stratified by age groups and sex. BMI categories are shown as underweight (blue), healthy weight (green), overweight (yellow), and obese (red).

Table 5.2. Distribution and association between age category and body mass index (BMI) category by sex

	18-24 Years		25-44 Years		45-54 Years		X^2	Cramér's V
BMI Category	**N**	**%**	**N**	**%**	**N**	**%**		
Female[a]							93.19	0.15
Underweight	6	0.78	7	0.63	0	0.00		
Healthy Weight	449	58.62	460	41.33	27	25.47		
Overweight	272	35.51	518	46.54	56	52.83		
Obese	39	5.09	128	11.50	23	21.70		
Male[a]							291.47	0.19
Underweight	1	0.07	8	0.33	0	0.00		
Healthy Weight	554	40.80	472	19.67	36	11.04		
Overweight	588	43.30	1158	48.27	158	48.47		
Obese	215	15.83	761	31.72	132	40.49		

[a] $p < 0.001$

Figure 5.2 and Table 5.2 Summary

Figure 5.2 and Table 5.2 show a clear association between age category and BMI category for both males and females. The stacked bar charts in Figure 5.2 highlight a visible shift in BMI category distribution status across age groups, with the percentage of subjects classified as healthy weight decreasing and the percentage of subjects classified as overweight and obese increasing with age for both sexes. The distributional differences of BMI categories across groups were statistically significant for both females ($\chi 2 = 93.13$, $p < 0.001$) and males ($\chi 2 = 291.47$, $p < 0.001$). However, the Cramér's V values (female = 0.15; male = 0.19) indicate the magnitude of the association is weak.[6]

Analyses Comparison

The continuous and categorical analyses between BMI and age across sexes showed the same general pattern but emphasized different aspects of the relationship. The continuous analysis highlights a clearer trend in how BMI *gradually* increases with age for both sexes and provided a clear estimate of how much BMI increases per year. However, the categorical analysis made the practical relationships and implications easier to see. The stacked bar charts in Figure 5.2 showed a clear visible shift from healthy weight categories toward overweight and obese categories across age groups. They also more clearly revealed the males had a smaller proportion of subjects classified in the healthy weight category in the youngest age group, even before considering age-related increases. This distinction made it easier to recognize how other differences (such greater muscle mass among males that may influence BMI) could also be important to consider. Overall, the continuous analyses had the greatest utility in providing a precise estimate of the relationship between variables, while the categorical analysis had the greatest utility in illustrating how weight status actually differs across age groups and between sexes.

References

1. Paquette SP, Parham JL. 2012 US Army Anthropometric Working Databases (ANSUR II). Natick, MA: US Army Natick Soldier Research, Development and Engineering Center; 2017.
2. Hotzman J, Gordon CC, Bradtmiller B, Corner BD, Mucher M, Kristensen S, Paquette S, Blackwell CL. Measurer's Handbook: US Army and Marine Corps Anthropometric Surveys, 2010–2011. Natick, MA: US Army Natick Soldier Research, Development and Engineering Center; 2011.
3. U.S. Army. Requirements to Join. https://www.goarmy.com/how-to-join/requirements. Accessed February 16, 2026.
4. HOKA. Sizing Information. https://www.hoka.com/en/us/sizing-information.html. Accessed February 20, 2026.
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6. Sapra, RL, Saluja, S. Understanding statistical association and correlation. *CMRP*. 2021;11(1):31-38. doi: 10.4103/cmrp.cmrp_62_20

Appendix

Do File Content

```
*Author: Savannah Daines
*Purpose: DMAV course project 1

cd "/Users/savannahdaines/Desktop/DMAP/Assignments/Project 1/Datasets"


************************** SECTION 1 **********************************

*See word document assignment report for introduction text



************************** SECTION 2 **********************************


******** Labeling

* Label Dataset
label data "ANSUR II exploratory data analysis dataset"

* Label Variables and Variable Levels according to MRF pdf:

** Label categorical labels
label variable subjectnumericrace "Self-reported Race(s)
label define subjectnumericrace_lbl 1 "White" 2 "Black" 3 "Hispanic" 4
"Asian" 5 "Native American" 6 "Pacific Islander" 8 "Other"
label values subjectnumericrace subjectnumericrace_lbl

label variable dodrace "Department of Defense Self-reported Single Race"
label variable subjectnumericrace "Self-reported Race(s)"
label define dodrace_lbl 1 "White" 2 "Black" 3 "Hispanic" 4 "Asian" 5 "Native
American" 6 "Pacific Islander" 8 "Other"
label values dodrace dodrace_lbl

** Label numerical labels
label variable ethnicity "Self-reported ethnicity"
label variable gender "Binary Gender Category"
label variable age "Age (years)"
label variable component "Component of the Army"
label variable branch "Branch of the Army"
label variable writingpreference "Preferred Writing Hand"
label variable installation "U.S. Army Installation Location of Measurement"
label variable test_date "Date of Measurement Tests"
label variable weightlbs "Self Reported Weight (lbs)"
label variable heightin "Self-reported Height (in)"
```

```
label variable weightlbs "Self-reported Weight (lbs)"
label variable thumbtipreach "Thumbtip Reach (mm)"
label variable span "Span (mm)"
label variable footlength "Foot Length (mm)"
label variable kneeheightmidpatella "Knee Height, Midpatella (mm)"
label variable waistheightomphalion "Waist Height (Omphalion) (mm)"
label variable functionalleglength "Functional Leg Length (mm)"
label variable cervicaleheight "Cervical Height (mm)"
label variable trochanterionheight "Trochanterion Height (mm)"
label variable stature "Stature (mm)"
label variable waistcircumference "Waist Circumference (Omphalion) (mm)"
label variable chestcircumference "Chest Circumference (mm)"
label variable bicristalbreadth "Bicristal Breadth (mm)"
label variable hipbreadth "Hip Breadth (mm)"
label variable hipbreadthsitting "Hip Breadth Sitting (mm)"
label variable weightkg "Weight (kg)"
label variable date "Date of Measurement"
label variable strdate "Date of Measurement"

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v1.dta"




******** Initial exploration

codebook // explore variables

tab1 subjectnumericrace dodrace ethnicity gender branch writingpreference
installation /// one way tabs of categorical variables


sum age weightlbs heightin thumbtipreach span footlength kneeheightmidpatella
waistheightomphalion functionalleglength cervicaleheight trochanterionheight
stature waistcircumference chestcircumference bicristalbreadth hipbreadth
hipbreadthsitting weightkg, detail /// look at percentiles & distribution of
continous variables

/*Notes on findings from initial exploration:
-
- Some extrme outliers in weightkg
- self reported weight has some 0's
- Age range includes 17
- subjectnumericrace has coding errors - numbers outside of 1-8 that are not
one of the 3 negatives
- Ethnicity has a lot of missing
- Writing preference has some inconsistencies in the string entries
- Date has a lot of missing
```

```
- Some extreme outliers in thumbtipreach*/



******** Unique key creation

gen id = _n // assign unique sequential number to each observation

label variable id "Unique Observation ID"   // add label

isid id // check uniquness and all are unique

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v2.dta" // save



******** Recdoing missing values

mvdecode _all, mv(-77=.a \ -88=.b \ -99=.c) // change the negative values to
special missing for all numeric variables; used AI (ChatGPT) to assist with
this code identification

/* Key - per the assignment instructions
.a = -77 = not recorded
.b = - 88 = refused measurement
.c = -99 = unknown missing */

summarize, detail // check the smallest number to see if these negatives got
coded right; everything looks good

* Add labels to the special missing values, reasons for missing are defined
by the assignment instructions and with details listed above in the key

label define misslbl ///
.a "Not Recorded" ///
.b "Refused Measurement" ///
.c "Unknown Missing"

foreach var of varlist _all {
    capture confirm numeric variable `var'
    if !_rc {
        label values `var' misslbl
    }
} // attach the labels
```

```
tab stature, missing // check to make sure labels were added; used AI
(ChatGPT) to assist with this code identification

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v3.dta"



******** Change units of measure

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v4_mm_measures.dta"

* MM measures to cm
** Conversion loop
foreach var in thumbtipreach span footlength kneeheightmidpatella
waistheightomphalion functionalleglength cervicaleheight trochanterionheight
stature waistcircumference chestcircumference bicristalbreadth hipbreadth
hipbreadthsitting {
    replace `var' = `var' / 10 if `var' < .
} // changes all the variables from mm to cm

tab stature, missing // check to special missing values; used AI (ChatGPT) to
assist with this code identification

** Fix all the labels to account for change to cm
label variable thumbtipreach "Thumbtip Reach (cm)"
label variable span "Span (cm)"
label variable footlength "Foot Length (cm)"
label variable kneeheightmidpatella "Knee Height, Midpatella (cm)"
label variable waistheightomphalion "Waist Height (Omphalion) (cm)"
label variable functionalleglength "Functional Leg Length (cm)"
label variable cervicaleheight "Cervical Height (cm)"
label variable trochanterionheight "Trochanterion Height (cm)"
label variable stature "Stature (cm)"
label variable waistcircumference "Waist Circumference (Omphalion) (cm)"
label variable chestcircumference "Chest Circumference (cm)"
label variable bicristalbreadth "Bicristal Breadth (cm)"
label variable hipbreadth "Hip Breadth (cm)"
label variable hipbreadthsitting "Hip Breadth Sitting (cm)"
label variable weightkg "Measured Weight (kg)"

* Inches to cm
replace heightin = heightin * 2.54 if heightin < . // change from in to cm //
used 2.54 cm is 1 inch conversion
rename heightin heightcm // rename
label variable heightcm "Self-reported Height (cm)" // relabel

tab heightcm, missing  // check missing
```

```
sum // check to make sure conversion worked

* Lbs to kg
replace weightlbs = weightlbs / 2.205 if weightlbs < . // change from lbs to
kg // used 2.205 kg is 1 lb conversion factor
rename weightlbs weightkg_selfreport // rename
label variable weightkg_selfreport "Self-reported Weight (kg)" // relabel

sum // check to make sure conversion worked


save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v5.dta"




******** Inspecting all variables for suspect/unreasonable values and
addressing them with either special missing values or flag

* Special Missing Values (SMV) Key

/*
.a = -77 = not recorded
.b = - 88 = refused measurement
.c = -99 = unknown missing
.d = value is a zero, which is biologically impossible
.e = suspiciously small (but not a zero) - see cutpoints for specific
variables
.f = suspiciously large - see cutpoints for specific variables */

* Additions to SMV

label define misslbl ///
.d "Value Recorded as 0" ///
.e "Suspiciously Small Non-Zero" ///
.f "Suspiciously large", add // add labels to the new special missing values


save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v6.dta"


* Variable inspection

***** AGE *****
```

```
sum age, detail // looks good but wil need to drop the underage subjects
later
tab age  // looks good but wil need to drop the underage subjects later


***** SELF-REPORTED WEIGHT (KG) *****

sum weightkg_selfreport, detail // inspect distribution
recode weightkg_selfreport (0=.d) // address zeros
tab weightkg_selfreport, missing // check
graph box weightkg_selfreport
sum weightkg_selfreport, detail

* All remaining values are biologically plausible, though there are some
suspicious values that will be flagged. Army enlistment standards are used as
contextual cutoffs for flagging per Army website numbers.
(https://www.goarmy.com/how-to-join/requirements). Additionally, soldiers are
assessed every 6 months with a weight screening per this document
(https://www.army.mil/e2/downloads/rv7/r2/policydocs/r600_9.pdf). Because
none of the weights are biologically impossible and there could be some
underlying medical issue driving a signficant weight change and are also self
reported, this supports an argument for flags rather than recoding to
missing.
* Per Army website:
* Maximum weight for tallest/oldest males = 250 lbs (113.398 kg)
* Maximum weight for tallest/oldest females = 236 lbs (107.048 kg)
* Minimum weight = 91 lbs (41.2769 kg) for both genders.

* Flag creation
gen flag_weightkg_selfreport = 0 // create flag variable and start everything
at 0

replace flag_weightkg_selfreport = 1 if weightkg_selfreport < . & ///
((gender == "Male" & (weightkg_selfreport > 113.398 | weightkg_selfreport <
41.2769)) | ///
(gender == "Female" & (weightkg_selfreport > 107.048 | weightkg_selfreport <
41.2769)))
replace flag_weightkg_selfreport = . if missing(weightkg_selfreport) // I
decided that if smv are retained in the original, then just have a generic
missing is okay for the flag

tab flag_weightkg_selfreport, missing // check flag distribution

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v7.dta", replace


*****  SELF-REPORTED HEIGHT (CM) *****
```

```
sum heightcm, detail  // inspect distribution

* Per Army enlistment standards (https://www.goarmy.com/how-to-
join/requirements):
* Minimum height for both  male/female = 58 inches (147.32 cm)
* Maximum height for both male/female = 80 inches (203.2 cm)

/*Based on the count ouputs comparing self reported vs. measured height as
well as the distribution of the data, it could be possible that someone made
a typo in recording the self reported height, they were considering height
with shoes on or off, lack basic common sense about what is a realistic
height, or maybe have some kind of injury like a concussion that is impairing
their reasoning ability. Since this is a self reported measure, no values
will changed to missing. Instead, a flag variable using the website's height
ranges will be created to make note of these values. */

count if heightcm > 203.2 & heightcm < . // equals 7
count if (heightcm > 203.2 & heightcm < .) & (stature > 203.2 & stature < .)
// equals 0

* Flag creation
gen flag_heightcm = 0 // create flag variable and start everything at 0
replace flag_heightcm = 1 if (heightcm < . ) & (heightcm > 203.2 | heightcm <
147.32)
replace flag_heightcm = . if missing(heightcm)

tab flag_heightcm, missing // check flag distribution

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v8.dta"


*****  THUMBTIP REACH (CM) *****

sum thumbtipreach, detail // check out distrubition
graph box thumbtipreach //

* There are some biologically implausible values that should be changed to
missing.

replace thumbtipreach = .f if (thumbtipreach > 150  & thumbtipreach < .) //
replace with missing value with the cut off for being biologically too large
and also considered the distrubution with extreme outliers

tab thumbtipreach, missing // check work

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v9.dta"
```

```
***** SPAN (CM) *****

sum span, detail // look at distribution
graph box span

sum stature, detail // look at distribution of stature and compare it to span
since the two measures are typically very close to each other.

* Everything seems biologiclaly plasuable. No flags/special missing values
needed


*****  Footlength (CM) *****

sum footlength, detail // look at distribution

summarize footlength if gender == "Male", detail // look at distribution by
gender
graph box footlength if gender == "Male"
summarize footlength if gender == "Female", detail // look at distribution by
gender
graph box footlength if gender == "Female"

* There are some smaller and larger foot lengths (for females the range
length equates to about size 3/4 womens shoe to size 12/13 and for males size
3/4 to size 15/16 on various brand sizing charts). There could be some kind
of data entry error but they are not biologically impossible. Therefore,
values outside of common retail size ranges will be flagged but not recoded
to missing.

* Hoka shoe sizing was used for determining common retail sizes and foot
length (https://www.hoka.com/en/us/sizing-information.html). For men, this
was below a size 5 (23 cm foot length) or greater than size 15 (31.5 cm foot
length). For wommen, this was below a size 5 (22 cm foot length) or greater
than size 12 (27.9 cm foot length). The size 5 and 12/15 cut off was made
because that is the size range of Hoka shoes and also fit well with the
distrubution.

* Flag creation
gen flag_footlength = 0 // create flag variale

replace flag_footlength = 1 if footlength < . & ///
((gender == "Male" & (footlength > 31.5 | footlength < 23)) | ///
(gender == "Female" & (footlength > 27.9 | footlength < 22)))
replace flag_footlength = . if missing(footlength)

tab flag_footlength, missing // check flag distribution
```

```
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v10.dta"


***** STATURE (CM) *****

sum stature, detail

*Per the Army website (https://www.goarmy.com/how-to-join/requirements) 58
inches (147.32 cm) and 80 inches (203.2 cm) are the max and min heights for
both males and females. There are no values above the max. There are 15
values below the min. It could be possible that these subjects had some kind
of injury that decreased their stature, so these will be flagged with a flag
variable.

count if stature < 147.32
count if (stature > 203.2 & stature < .)

* Flag creation
gen flag_stature = 0 // create flag
replace flag_stature = 1 if stature < 147.32
replace flag_stature = . if missing(stature)

tab flag_stature, missing // flag numbers check out

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v11.dta"


*****  kneeheightmidpatella (CM) *****

sum kneeheightmidpatella, detail  // look at distribution
graph box kneeheightmidpatella

twoway scatter kneeheightmidpatella stature // look at relative to stature

* compared min and max as a proportion to the min and max of stature and it
seems biologically reasonable

* no concerns for suspicious values here


*****  waistheightomphalion (CM)*****

sum waistheightomphalion, detail // look at distribution
graph box waistheightomphalion
twoway scatter waistheightomphalion stature // look at relative to stature
```

```
* Compared min and max as a proportion to the min and max of stature and it
seems biologically reasonable

* no concerns for suspicious values here


*****  functionalleglength (CM) *****

sum functionalleglength, detail // look at distribution
graph box functionalleglength
twoway scatter functionalleglength stature // look at relative to stature

* Also compared min and max as a proportion to the min and max of stature and
it seems biologically reasonable

* no concerns for suspicious values here


****** cervicaleheight (CM) ********

sum cervicaleheight, detail // look at distribution
graph box cervicaleheight
twoway scatter functionalleglength stature // look at relative to stature

* Also compared min and max as a proportion to the min and max of stature and
it seems biologically reasonable

* no concerns for suspicious values here


*****  trochanterionheight (CM) *****

sum trochanterionheight, detail // look at distribution
graph box trochanterionheight
twoway scatter functionalleglength stature // look at relative to stature

* compared min and max as a proportion to the min and max of stature and it
seems biologically reasonable

* no concerns for suspicious values here


*****  waistcircumference (CM) *****

summ waistcircumference, detail // check distrubition

summ waistcircumference if gender == "Male", detail // check distrubition by
gender
summ waistcircumference if gender == "Female", detail
```

```
twoway scatter waistcircumference  weightkg_selfreport // look to see if
there is anything substantially deviated from a connection to waist to also
explore suspicous values

graph box waistcircumference if gender == "Male"
graph box waistcircumference if gender == "Female"

* All values are biologically plausible, but there a some larger values that
would likely not meet body composition standards but they could still be
possibile for circumstances such as being temporarily noncompliant or
medicallly exempt. Therefore, flag variables will be created based on
cutpoints for the 1st and 99th percentile for both genders.

* Flag creation

gen flag_waistcircumference = 0 // create flag variable and start everything
at 0

replace flag_waistcircumference = 1 if waistcircumference < . & ///
((gender == "Male" & (waistcircumference > 121.4 | waistcircumference <
72.5)) | ///
(gender == "Female" & (waistcircumference > 111.3 | waistcircumference <
66.5)))
replace flag_waistcircumference = . if missing(waistcircumference)

tab flag_waistcircumference, missing // check flag distribution

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v13.dta"


*****  chestcircumference (CM) *****

summ chestcircumference if gender == "Male", detail  // check distrubition
summ chestcircumference if gender == "Female", detail  // check distrubition

* All values are biologically plausible. There are some that could suggest a
very slim individual or possibly masectomy or extreme amounts of muscle +
leanness (to still meet body composition requirements), though it could still
be possible. Therefore, no flags will be created.


*****  bicristalbreadth (CM) *****

summ bicristalbreadth if gender == "Male", detail  // check distrubition
summ bicristalbreadth if gender == "Female", detail  // check distrubition
```

```
* All values are biologically plausible. There are some extremes, but still
realistic possibibilities. Therefore, no flags will be created.


*****  hipbreadth (CM) *****

summ hipbreadth if gender == "Male", detail  // check distrubition
summ hipbreadth if gender == "Female", detail  // check distrubition

* All values are biologically plausible. There are some extremes, but still
realistic possibibilities. Therefore, no flags will be created.


****** hipbreadthsitting (CM) ********

summ hipbreadthsitting if gender == "Male", detail  // check distrubition
summ hipbreadthsitting if gender == "Female", detail  // check distrubition

* All values are biologically plausible. There are some extremes, but still
realistic possibibilities. Therefore, no flags will be created.


*****  weightkg (CM) *****

summ weightkg if gender == "Male", detail  // check distrubition
summ weightkg if gender == "Female", detail  // check distrubition

* Explore some high values
count if weightkg > 400 & weightkg < . // count is 39
count if weightkg > 200 & weightkg < . // count is 39
count if weightkg > 150 & weightkg < . // count is 39

** The count if 39 for all three of the kg checks above. It is biologically
implausible that someone weights more tahn 400 kg. Therefore, these values
will be set as special missing values.

* Set to missing
replace weightkg = .f if (weightkg > 400  & weightkg < .) // replace with
missing value with the cut off for being biologically too large

tab weightkg, missing // check work

summ weightkg if gender == "Male", detail  // check distrubition
summ weightkg if gender == "Female", detail  // check distrubition

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v14.dta"
```

```
*****  subjectnumericrace *****

tab subjectnumericrace // the pdf manual reports there should only be races
1-8. There are numerous races listed as greater than. These will recoded to
smv.

***Special Missing Values Key ***

/*
.a = -77 = not recorded
.b = - 88 = refused measurement
.c = -99 = unknown missing
.d = value is a zero, which is biologically impossible
.e = suspiciously small (but not a zero) - see cutpoints for specific
variables
.f = suspiciously large - see cutpoints for specific variables
.g = invalid value number per pdf manual */

* Additions to the key and labeling

label define misslbl ///
.g "Invalid Value per pdf Manual" , add /// add labels to the new special
missing values

replace subjectnumericrace = .g if subjectnumericrace > 8 &
subjectnumericrace < . // create the smv
tab subjectnumericrace, missing // check

label define subjectnumericrace_lbl 1 "White" 2 "Black" 3 "Hispanic" 4
"Asian" 5 "Native American" 6 "Pacific Islander" 8 "Other" /// add labels

label values subjectnumericrace subjectnumericrace_lbl /// add labels

tab subjectnumericrace, missing // checked

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v16.dta"


****** dodrace ******

tab dodrace, missing /// check for missing/weird things

label define dodrace_lbl 1 "White" 2 "Black" 3 "Hispanic" 4 "Asian" 5 "Native
American" 6 "Pacific Islander" 8 "Other" /// add labels

label values dodrace dodrace_lbl /// add labels

tab dodrace, missing /// check for missing/weird things
```

```
replace dodrace = .g if dodrace == 7 // 7 is not a valid option per pdf

tab dodrace, missing // checked

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v17.dta"


****** ethnicity ******

tab ethnicity, missing /// check for missing/weird things

* Address potential spaces or capitlization issues
replace ethnicity = trim(ethnicity) // remove trailing/leading spaces if
there; used CHATGPT to identify code
replace ethnicity = lower(ethnicity) // put all in lowercase; used CHATGPT to
identify code

tab ethnicity, missing // check for weird stuff like unknown, n/a, numbers,
nonsensical responses, blanks, etc

count if ethnicity == "" // count blanks // numerous found

* Create a flag for blanks
gen flag_ethnicity_missing = 0 // create flag for missing ethncity since no
smv for string
replace flag_ethnicity_missing = 1 if ethnicity == ""

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v18.dta"


****** gender ******

tab gender, missing // looks good


****** component ******

tab component, missing // looks good


****** branch ******

tab branch, missing // looks good


****** writingpreference ******
```

```
tab writingpreference, missing

* per the pdf options include Writing Preference; "Right hand", "Left hand",
or "Either hand (No preference)" . Therefore, the 25 "Either han" will be
replaced with "Either hand (No preference)" per below:

replace writingpreference = "Either hand (No preference)" if
writingpreference == "Either han"

tab writingpreference, missing // check

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v19.dta"


****** installation ******

tab installation, missing // looks good


****** test_date ******

tab test_date, missing // looks good, all within correct dates


****** date ******

tab date, missing

* there are many missing values recorded as . and will be changed to smv

***Special Missing Values Key ***

/*
.a = -77 = not recorded
.b = - 88 = refused measurement
.c = -99 = unknown missing
.d = value is a zero, which is biologically impossible
.e = suspiciously small (but not a zero) - see cutpoints for specific
variables
.f = suspiciously large - see cutpoints for specific variables
.g = invalid value number per pdf manual
.h = recorded as . in the dataset */

*Additions to key

label define misslbl ///
```

```
.h "Recorded as . in the Dataset", add /// add labels to the new special
missing values

replace date = .h if date == .

tab date, missing // check work


****** strdate ******

tab strdate, missing // numerous blanks exist

gen flag_strdate_missing = 0 // create flag for missing ethncity since no smv
for string
replace flag_strdate_missing = 1 if strdate == ""


* Label all the new flag varibles

label variable flag_weightkg_selfreport "Suspicious Self-reported Weight
(kg)"
label variable flag_heightcm "Suspicious Self-reported Height (cm)"
label variable flag_footlength "Suspicious Foot Length (cm)"
label variable flag_stature "Suspicious Stature (cm)"
label variable flag_waistcircumference "Suspicious Waist Circumference
(Omphalion) (cm)"
label variable flag_ethnicity_missing "Flag for Blank Self-reported Ethnicity
"
label variable flag_strdate_missing "Flag for Blank String Date"

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v20.dta"


* Remove underage

** The military approval for this evaluation allowed only those who were of
legal age (18) to participate

count if age < 18 // there are 2 subjects underage
drop if age < 18 // drop
count if age < 18  // check

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v21.dta"
```

```
******** Identify and address duplicates (part one) *****

duplicates report // check exact duplicates // there are none

* Explore possibilities of combinations of matching variables:

duplicates report component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch // output says there are
1842 observations that are duplicate pairs for these variables

duplicates report component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch date // output says there
are 0 observations that are duplicate pairs for these variables, which is the
same as above except for date. It would be extremely unlikely that these
would be different individuals based on this and it is likely they are the
same individuals but a different date of measurement. These will be flagged
and addressed below.

duplicates tag component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch, gen(flag_duplicates1) //
create flag for the duplicate situation above

tab flag_duplicates1 // check flag

label variable flag_duplicates1 "Flag for duplicates1"  // label flag

browse if flag_duplicates1 > 0 // explore what these duplicates look like
across other variables, good to continue


* Pull out the duplicates1
keep if flag_duplicates1 > 0 //  get just the duplicattes
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/duplicates1_records.dta" // save duplicates1 dataset

use "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v21.dta" // go back to full dataset

duplicates tag component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch, gen(flag_duplicates1) //
add flag again for the duplicate situation above

tab flag_duplicates1 // check flag

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v22.dta"

* Stata keeps the first observation in sort order, so we need to sort:
```

```
gsort component gender age stature weightkg waistcircumference hipbreadth
hipbreadthsitting installation branch -date // sort left to right/bring the
"identical" measures together, have the dated sorted in desecending order so
that the newest data is first; // used CHATGPT to help identify code

browse if flag_duplicates1 > 0 // check to see if sorting worked. Looks like
the "duplicates" have an .h

count if flag_duplicates1 > 0 & date == .h // this equals 921, which is
consistent with the 1842 duplicate observations above.

* Drop the observations with a .h for date so that that we keep the
observation with the most recent valid date. The valid date comes first in my
current sort order and no duplicate pair for this flag_duplicates1 have 2
valid dates. Therefore, I can proceed with dropping.

duplicates drop component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch, force // drop

browse if flag_duplicates1 > 0 // check work
count if flag_duplicates1 > 0 // check work - 921 so it looks good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v23.dta"



tab gender // check # of observations. We now have 6,108. The combined ansur2
dataset for males and females should have 6,068.


* Explore where the other duplicates might have come from:

** remove one variable at a time from above duplicate identification and get
counts

duplicates report component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch // 0

duplicates report component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation date // 0

duplicates report component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting branch date //

duplicates report component gender age stature weightkg waistcircumference
hipbreadth installation branch date // 0
```

```
duplicates report component gender age stature weightkg waistcircumference
hipbreadthsitting installation branch date // 0

duplicates report component gender age stature weightkg hipbreadth
hipbreadthsitting installation branch date // 0

duplicates report component gender age stature waistcircumference hipbreadth
hipbreadthsitting installation branch date // 0

duplicates report component gender age weightkg waistcircumference hipbreadth
hipbreadthsitting installation branch date // 0

duplicates report component age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch date // 0

duplicates report component gender age stature weightkg waistcircumference
hipbreadth hipbreadthsitting installation branch date //

duplicates report gender age stature weightkg waistcircumference hipbreadth
hipbreadthsitting installation branch // 0

*Explore other combinations of matches

duplicates report installation branch gender age stature weightkg date

duplicates report installation branch gender age stature

duplicates report installation branch gender age stature date

duplicates report installation branch gender age stature date footlength

** Possible duplicates will be relevaluated again if any potential patterns
in data that suggest duplicates are discovered.


******** Create BMI variables

* Create Continous BMI

** First we neeed to convert stature to meters. I will create a new variable
in m:

gen stature_m = . // create stature in m variable
replace stature_m = stature/100 if stature < .
summ stature stature_m // check work
tab stature, missing // check work
label variable stature_m "Stature (m)" // label
```

```
** bmi formula is weight in weight in kg/ height in meters squared. This
formula is applied below in the creation of the bmi variable.

gen bmi = . // create bmi variable
replace bmi = .i if (missing(weightkg) | missing(stature_m))
replace bmi = weightkg/(stature_m^2) if (weightkg < . & stature_m < .)
summ bmi, detail // check work
misstable summarize bmi // make sure missing was protected
label variable bmi "Body Mass Index" // label

* I need to have a smv for bmi if missing weightkg or missing stature_m:

***Special Missing Values Key ***

/*
.a = -77 = not recorded
.b = - 88 = refused measurement
.c = -99 = unknown missing
.d = value is a zero, which is biologically impossible
.e = suspiciously small (but not a zero) - see cutpoints for specific
variables
.f = suspiciously large - see cutpoints for specific variables
.g = invalid value number per pdf manual
.h = recorded as . in the dataset
.i = missing weightkg or missing stature_m for bmi variables */

* Additions to spv

label define misslbl ///
.i "Missing weightkg or missing stature_m for bmi variables", add /// add
labels to the new special missing values

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v24.dta"


** Create Categorical BMI

gen bmi_cat = . // creates new variable

replace bmi_cat = 1 if bmi <18.5 & bmi < .
replace bmi_cat = 2 if bmi >=18.5 & bmi <25
replace bmi_cat = 3 if bmi >=25 & bmi <30
replace bmi_cat = 4 if bmi >=30 & bmi < . // protects missing values
replace bmi_cat = .i if bmi ==.i // missing weightkg or missing stature_m

tab bmi_cat, missing //  looks good

* Add labels
```

```
label variable bmi_cat "Body Mass Index Category Classifications" // label
variable

label define bmi_cat_lbl 1 "Underweight" 2 "Healthy Weight" 3 "Overweight" 4
"Obese" .i "missing weightkg or missing stature_m" // creates label set
label values bmi_cat bmi_cat_lbl  // attaches label set to variable
tab bmi_cat, missing // check labels // all labels look good
describe bmi_cat // check labels // all labels look good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v25.dta"


** Check BMI for suspects and potentially needing flags

summ bmi, detail //
histogram bmi // none of the values are biologically implausiable. Some are
potentially really thin at a level of some kind of medical or mental health
condition, but still possible. There are also some larger bmi values, which
considering soldiers may be much more muscular, they are not overly
suspicious. Therefore, flags will be for those that are underweight (BMI less
than 18.5) and in the Obese II or greater category (BMI greater than 35).
This appears to align well with the histogram distribution for flagging
extreme values.

gen flag_bmi = 0 // create flag variable
replace flag_bmi = 1 if (bmi < 18.5 | bmi > 35) & bmi < .
replace flag_bmi = . if bmi >= .
label variable flag_bmi "Suspicious BMI"

browse if flag_bmi > 0 // check looks good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v26.dta"



******** Identify and address duplicates (part two) *****

* in browsing the dataset with the flag_bmi > 0, I noticed there may be some
potentially duplicates with one observation have a suspiously large weightkg.
These also had an .h for the date in the large value.

* Tested various combinations of measurements (direclty measured) and
variables that should not change if duplicates. The combination included all
direclty measured variables with the exception of weight. This is the
duplicates report with the variables determined to reflect duplication
(leaves out weightkg)
```

```
duplicates report gender age component branch installation thumbtipreach span
footlength waistheightomphalion functionalleglength cervicaleheight
trochanterionheight stature waistcircumference chestcircumference
bicristalbreadth hipbreadth hipbreadthsitting

**  Tag these duplicates
duplicates tag gender age component branch installation thumbtipreach span
footlength waistheightomphalion functionalleglength cervicaleheight
trochanterionheight stature waistcircumference chestcircumference
bicristalbreadth hipbreadth hipbreadthsitting, generate(dup2)

rename dup2 flag_duplicates_weighterror // rename

**  Sort the dataset to for how I would go about dealing with the weightkg
duplicates and also include date in desending order since I noticed the .h in
the suspect weightkg value.

gsort gender component branch installation thumbtipreach span footlength
waistheightomphalion functionalleglength cervicaleheight trochanterionheight
stature waistcircumference chestcircumference bicristalbreadth hipbreadth
hipbreadthsitting age kneeheightmidpatella weightkg -date // used CHATGPT to
help identify code

browse if flag_duplicates_weighterror > 0

count if flag_duplicates_weighterror > 0
count if flag_duplicates_weighterror > 0 & date == .h // tsting my theory of
the .h with dates and the suspect weightkg values
count if flag_duplicates_weighterror > 0 & weightkg == .f
count if flag_duplicates_weighterror > 0 & date == .h & weightkg == .f

** Based on the matching measured anthropometrics, browsing,  and counts,
these are duplicates that should be pulled out.

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v27.dta"

** Pull out the flag_duplicates_weighterror
keep if flag_duplicates_weighterror > 0 //  get just the duplicattes
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/duplicates_weighterror_records.dta" // save duplicates_weighterror
dataset

use "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v27.dta" //  go back to full dataset

* Stata keeps the first observation in sort order.
```

```
gsort gender age component branch installation thumbtipreach span footlength
waistheightomphalion functionalleglength cervicaleheight trochanterionheight
stature waistcircumference chestcircumference bicristalbreadth hipbreadth
hipbreadthsitting -weightkg // sort left to right/bring the "identical"
measures together, have the weight sorted in desecending order so that the
valid weight is first; used CHATGPT to help identify code

browse if flag_duplicates_weighterror > 0 // check to see if sorting worked.

count if flag_duplicates_weighterror > 0 & weightkg == .f // this equals 39,
which is consistent with my assessment above

* I want to drop the ones with observations with a .f for weightkg so that
that we keep the observation with not suspicous weightkg and also a valid
date

duplicates drop gender age component branch installation thumbtipreach span
footlength waistheightomphalion functionalleglength cervicaleheight
trochanterionheight stature waistcircumference chestcircumference
bicristalbreadth hipbreadth hipbreadthsitting, force // drop

browse if flag_duplicates_weighterror > 0 // check work - valid date there
count if flag_duplicates_weighterror > 0 // check work 39 - so it looks good

label variable flag_duplicates_weighterror "duplicates due to suspicious
weightkg"

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v28.dta"

** Look for any remaining duplicates

tab gender, missing // there are 6,069 observations. This means we have one
duplicate based on the orginal dataset numbers.

* Explore possibilities:

tab date, missing /// there are 3 observations with .h (meaning this was
recorded as . in the dataset ; because the other identified duplicates also
have this .h issue, these should be explored in more detail to see if the
extra observation can be identified).

browse if date == .h // there is one observation that is missing all measured
anthrometrics. This is subject ID 6516.

browse if gender == "Male" & component == "Regular Army" & branch == "Combat
Arms" & writingpreference == "Right hand" & installation == "Fort Blis" & age
== 44 & dodrace == "White" & subjectnumericrace == "White" // these are the
datapoints (with the exception of self reported height and weight) we have on
```

subject ID 6516. I browsed the dataset to look if there are any subjects with these values that self reported the same height and weight. There is one subject that fits this, which is subject 825. Therefore, I conclude that this subject 6516 is the remaining duplicate measure and will be pulled out as a duplicate because it has no (incomplete) anthrometric measures.

```
* Create flag for this duplicate.
gen flag_duplicates_incomplete = 0 // create flag variable and start
everything at 0
replace flag_duplicates_incomplete = 1 if (id == 6516) | (id == 825)
browse if flag_duplicates_incomplete > 0 // check to make sure they are right
ones

* duplicate dropping pre save
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v29.dta"

*Pull out the flag_duplicates_incomplete
keep if flag_duplicates_incomplete > 0 //  get just the duplicattes
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/duplicates_incomplete.dta" // save duplicates_incomplete dataset

use "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v29.dta" //  go back to full dataset

*drop the duplicate
drop if id == 6516 // drop the incomplete duplicate
browse if flag_duplicates_incomplete > 0 // check to make sure it got dropped
right

label variable flag_duplicates_incomplete "Flag for last duplicate with an
incomplete measurement assessment"

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v30.dta"

tab gender, missing /// we are at 6,068 observations, which is what we should
have based on the offical ANSURII pdf documentation!


******** Check bmi_cat for suspects and potentially needing flags

tab bmi_cat, missing


*Apply the smae flags from the bmi continous variable since those should be
the same.

gen flag_bmi_cat = 0 // create flag variable
```

```
replace flag_bmi_cat = 1 if (bmi < 18.5 | bmi > 35) & bmi < .
replace flag_bmi_cat = . if bmi >= .
label variable flag_bmi_cat "Suspicious BMI Category (based on BMI number)"

count if if flag_bmi > 0
count if if bmi_cat > 0 // both the categorical and continuous match in
counts
browse if flag_bmi_cat > 0 // check looks good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v31.dta"


******** Create Season Variable

* Create a variable that indicates the season the measurements were made.

**look at date variables to figure out an approach.
describe date // long storge,  %tdD_m_Y
tab date, missing // 2 missing values (.h)
describe strdate // 2 blank

*extract month from date; since date is already in a stata daily date format
no conversion is needed.

gen month = month(date)

replace month = .h if missing(date)

tab month, missing // looks good

label variable month "month"

*create season variable and add labels

gen season = .

replace season = 1 if inlist(month, 12, 1, 2)
replace season = 2 if inlist(month, 3, 4, 5)
replace season = 3 if inlist(month, 6, 7, 8)
replace season = 4 if inlist(month, 9, 10, 11)
replace season = .h if missing(date) // used ChatGPT to help identify code

label define season_lbl ///
1 "Winter" ///
2 "Spring" ///
3 "Summer" ///
4 "Fall" ///
.h "Missing date"
```

```
label values season season_lbl
label variable season "Season of Measurement"


tab season, missing // looks good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v32.dta"



******** Create Numerical Categorical Variable for Gender

*look at variables to figure out an approach.
describe gender // string
tab gender, missing //1985 female, 4083 male

*change from string to numeric categorical
encode gender, generate(gender_cat) label(gender_catlbl) // used ChatGPT to
help identify code

rename gender_cat gender_num_cat // rename

label variable gender_num_cat "Binary Gender Numeric Categorical" // label

summ gender_num_cat, detail // checked to make sure just ones and twos there
as
tab gender_num_cat, missing // labels are there, the male and female total
match the string gender variable numbers. No flags or suspect things.

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v33.dta"



***** Create Numerical Categorical Variable for Preferred Hand

*look at variables to figure out an approach
describe writingpreference // string

tab writingpreference, missing // 62 either, 656 left hand, 5350 right hand

*change from string to numeric categorical
encode writingpreference, generate(writingpreference_num_cat)
label(writingpreference_num_cat) // used ChatGPT to help identify code

label variable writingpreference_num_cat "Preferred Writing Hand Numeric
Categorical" // label
```

```
summ writingpreference_num_cat, detail // checked to make sure there are just
1-3 as there should be
tab writingpreference_num_cat, missing // labels are there, totals match the
string numbers. No flags or suspect things.

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v34.dta"



***** Body Type Categories creation

*Explore distrubtions of anthropometrics across males and females.

sum thumbtipreach footlength kneeheightmidpatella waistheightomphalion
functionalleglength cervicaleheight trochanterionheight stature
waistcircumference chestcircumference bicristalbreadth hipbreadth
hipbreadthsitting weightkg if gender_num_cat == 1, detail // female
distrubtions

sum thumbtipreach footlength kneeheightmidpatella waistheightomphalion
functionalleglength cervicaleheight trochanterionheight stature
waistcircumference chestcircumference bicristalbreadth hipbreadth
hipbreadthsitting weightkg if gender_num_cat == 2, detail // male
distrubtions

/*Given the military context of the dataset, I determined it would be best to
come up with a classification that is provides meaningful information of body
structure types in terms of factors that would be important for military
operations. With this in mind, I created three overarching indexes for the
algorithm that focus on mass relative to height, vertical body proportions,
and body breadth/robustness index. These indexes will provide meaningful
information related to military factors like plane/vehicle clearance or
weight fit, clothing and other gear fit, mobility potential, fit within tight
spaces, capacity for carrying heavy things, and potential strength-mass
tradeoffs of different personnel that might be important for specific
missions. Rather than using general civilian references for determining
cutpoints for the different indexes, I used the tertiles for each gender
within the dataset to provide a more meaningful assessment/comparison in the
context of the army and get about the same number of subjects for each index.
Indexes were then used to categorize body types based on build and frame.

**** INDEXES

* Index 1: Mass Index
- Use MHR, which is a new variable created, representing Mass Height Ratio,
which is calulcated as weightkg / stature_m
```

```
- Classify into low, medium, and high based on tertiles for male and female
using 1, 2, 3 for low, medium, and high

* Index 2: Vertical Proportion Index
- Use LSR, which is a new variable created, representing Functional Leg
Length Stature Ratio, which is calulcated as functionalleglength / stature_m
- Classify into short-legged, balanced, and long-legged based on on tertiles
for male and female using 1, 2, 3 for short-legged, balanced, and long-legged

* Index 3: Robustness Index
- Use TSR, which is new variable created, represented Trunk Stature Ratio,
which is calculculated as waistcircumference / stature_m
- Classify into narrow, average, and broad based on on tertiles for male and
female using 1, 2, 3 for narrow, average, and broad */

**** CREATE INDEXES AND DETERMINE TERTILES

* Check to see if any missing values across variables
tab weightkg, missing
tab stature_m, missing
tab waistcircumference, missing
tab functionalleglength, missing // there are no missing values any of these
measures


* Create new variables

** mhr (see INDEXES section above for more info)
gen mhr = . // create variable
replace mhr = weightkg/(stature_m) if (weightkg < . & stature_m < .)
summ mhr, detail // check work
label variable mhr "Mass Height Ratio" // label

** lsr (see INDEXES section above for more info)
gen lsr = . // create variable
replace lsr = functionalleglength /(stature_m) if (functionalleglength < . &
stature_m < .)
summ lsr, detail // check work
label variable lsr "Functional Leg Length Stature Ratio" // label

** tsr (see INDEXES section above for more info)
gen tsr = . // create variable
replace tsr = waistcircumference/(stature_m) if (waistcircumference < . &
stature_m < .)
summ tsr, detail // check work
label variable tsr "Trunk Stature Ratio" // label

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v35.dta"
```

```
* Create tertile variables by sex

ssc install egenmore // install needed feature // identified by Chat GPT

** mhr

bysort gender_num_cat: egen mhr_tert = xtile(mhr), nq(3) // split data into
lowest third, middle third, and highest third values for each sex // used
ChatGPT to help identify code

label define mhr_tert_lbl 1 "Low" 2 "Medium" 3 "High" // add all the labels
label values mhr_tert mhr_tert_lbl
label var mhr_tert "mhr Tertile (Sex-specific)"

tab gender_num_cat mhr_tert, row // every looks good, tertiles are there and
there are 33.30-33.35% in each group.


** lsr

bysort gender_num_cat: egen lsr_tert = xtile(lsr), nq(3) // split data into
lowest third, middle third, and highest third values for each sex // used
ChatGPT to help identify code

label define lsr_tert_lbl 1 "Compact" 2 "Proportional" 3 "Elongated" // add
all the labels
label values lsr_tert lsr_tert_lbl
label var lsr_tert "lsr Tertile (Sex-specific)"

tab gender_num_cat lsr_tert, row // every looks good, tertiles are there and
there are 33.30-33.35% in each group.


** tsr

bysort gender_num_cat: egen tsr_tert = xtile(tsr), nq(3) // split data into
lowest third, middle third, and highest third values for each sex // used
ChatGPT to help identify code

label define tsr_tert_lbl 1 "Narrow" 2 "Average" 3 "Broad" // add all the
labels
label values tsr_tert tsr_tert_lbl
label var tsr_tert "tsr Tertile (Sex-specific)"

tab gender_num_cat mhr_tert, row // every looks good, tertiles are there and
there are 33.30-33.35% in each group.
```

```
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v36.dta"


**** Create the body types based on indexes and body type variable ****

/* I defined body types as based on 1) Build and 2) Vertical Proportion. I
determined to have 9 body types. Since there are 27 possible combinations and
the instructions say to have between 3-6 but Dr. Vanderslice said we could
have more if it was meaningful, this number struck an appropriate balance of
not collapsing into too few of groups to not have it be as meaningful/making
clear distinctions but also not going overboard in the number of groups.

1) Build was defined first and applied across all vertical proportion groups.
I derived build categories from
sex-specific tertiles of HMR and TSR as follows:

- Light: mhr_tert == 1 AND tsr_tert == 1
  (lowest tertile of mass and narrowest trunk relative to stature)

- Robust: mhr_tert == 3 OR tsr_tert == 3
  (highest tertile of mass and/or broadest trunk)

- Intermediate: all other combinations of mhr_tert and tsr_tert
  (neither Light nor Robust)

2) Vertical Proportion was defined using sex-specific tertiles of LSR as
follows:

- Compact: lsr_tert == 1
  (lower vertical proportion)

- Proportional: lsr_tert == 2
  (mid-range vertical proportion)

- Elongated:  lsr_tert == 3 → Elongated
  (higher vertical proportion)

Body type was then defined Vertical Proportion × Build combinations, which
created nine categories:

Compact-Light
Compact-Intermediate
Compact-Robust

Proportional-Light
Proportional-Intermediate
Proportional-Robust
```

```
Elongated-Light
Elongated-Intermediate
Elongated-Robust
*/

**** Create body types

* check to see if there are missing values
count if mhr_tert >= .
count if lsr_tert >= .
count if tsr_tert >= . // each of these 3 variables do not have missing
values, so I do not need to worry about protecting missing values as I create
body types.

** create the build variable

gen build = . // create variable

replace build = 1 if mhr_tert==1 & tsr_tert==1 // light
replace build = 3 if mhr_tert==3 | tsr_tert==3 // robust
replace build = 2 if build== . // intermediate (all other combinations)

***  Add labels
label define buildlbl ///
1 "Light" ///
2 "Intermediate" ///
3 "Robust"

label values build buildlbl
label variable build "Build (Light / Intermediate / Robust)"

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v37.dta"

** create body types and labels based on build and vertical proportion

gen bodytype = . // create variable

*compact (lsr_tert == 1)
replace bodytype = 1 if lsr_tert==1 & build==1 // Compact-Light
replace bodytype = 2 if lsr_tert==1 & build==2 // Compact-Intermediate
replace bodytype = 3 if lsr_tert==1 & build==3 // Compact-Robust

*proportional (lsr_tert == 2)
replace bodytype = 4 if lsr_tert==2 & build==1 // Proportional-Light
replace bodytype = 5 if lsr_tert==2 & build==2 // Proportional-Intermediate
replace bodytype = 6 if lsr_tert==2 & build==3 // Proportional-Robust

*elongated (lsr_tert == 3)
```

```
replace bodytype = 7 if lsr_tert==3 & build==1 // Elongated-Light
replace bodytype = 8 if lsr_tert==3 & build==2 // Elongated-Intermediate
replace bodytype = 9 if lsr_tert==3 & build==3 // Elongated-Robust

***  Add labels
label define bodylbl ///
1 "Compact-Light" ///
2 "Compact-Intermediate" ///
3 "Compact-Robust" ///
4 "Proportional-Light" ///
5 "Proportional-Intermediate" ///
6 "Proportional-Robust" ///
7 "Elongated-Light" ///
8 "Elongated-Intermediate" ///
9 "Elongated-Robust"

label values bodytype bodylbl
label variable bodytype "Body Type (Vertical Proportion x Build)"

** checks
count if missing(bodytype)
tab bodytype
tab gender_num_cat bodytype, row // everything looks good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v38.dta"


******** Categorical Unisex T Shirt Size

* Unisex 3600 t shirt size chart from Next Level Apparel
(https://www.nextlevelapparel.com/cdn/shop/files/3600-WEBSITE-SPEC_e289f037-
f085-4819-b229-eb662191ea49.pdf?v=9763116909218305737)

/* Size chart conversions:

1) Original flat chest width (inches)from website above:

XS  = 17.5
S   = 19
M   = 20.5
L   = 22
XL  = 24
2XL = 26
3XL = 28
4XL = 30
5XL = 32
6XL = 34
```

2) Conversion from flat width (half circumference) to full chest circumference. I multiplied by 2 to get the full chest circumference.

```
XS  = 17.5 × 2 = 35
S   = 19   × 2 = 38
M   = 20.5 × 2 = 41
L   = 22   × 2 = 44
XL  = 24   × 2 = 48
2XL = 26   × 2 = 52
3XL = 28   × 2 = 56
4XL = 30   × 2 = 60
5XL = 32   × 2 = 64
6XL = 34   × 2 = 68
```

3) Convert inches to cm to match database. 1 inch = 2.540 cm, so I multlpied by 2.540 and rounded to 3 decimal places.

Full chest circumference (cm):

```
XS  = 35 × 2.54 = 88.900 cm
S   = 38 × 2.54 = 96.520 cm
M   = 41 × 2.54 = 104.140 cm
L   = 44 × 2.54 = 111.760 cm
XL  = 48 × 2.54 = 121.920 cm
2XL = 52 × 2.54 = 132.080 cm
3XL = 56 × 2.54 = 142.240 cm
4XL = 60 × 2.54 = 152.400 cm
5XL = 64 × 2.54 = 162.560 cm
6XL = 68 × 2.54 = 172.720 cm
```

4) Logic for determining the best likely sizes: Give each subject the smallest shirt size that has a shirt full chest circumference that is greater to or equal their
measured chest circumference (shirt chest cirumference is not smaller than the subject's chest circumference). */

```
* Create the t shirt size variable

sum chestcircumference // the range is 69-146.9, so we only need to use sizes
XS to 4XL

count if chestcircumference >= . // there are no missing values, so I do not
need to worry about protecting missing values

gen tshirtsize = . // create variable

replace tshirtsize = 1 if chestcircumference <= 88.900
replace tshirtsize = 2 if chestcircumference > 88.900  & chestcircumference
<= 96.520
```

```
replace tshirtsize = 3 if chestcircumference > 96.520  & chestcircumference
<= 104.140
replace tshirtsize = 4 if chestcircumference > 104.140 & chestcircumference
<= 111.760
replace tshirtsize = 5 if chestcircumference > 111.760 & chestcircumference
<= 121.920
replace tshirtsize = 6 if chestcircumference > 121.920 & chestcircumference
<= 132.080
replace tshirtsize = 7 if chestcircumference > 132.080 & chestcircumference
<= 142.240
replace tshirtsize = 8 if chestcircumference > 142.240

* Add labels
label define shirtsizelbl ///
1 "XS" ///
2 "S" ///
3 "M" ///
4 "L" ///
5 "XL" ///
6 "2XL" ///
7 "3XL" ///
8 "4XL"

label values tshirtsize shirtsizelbl
label variable tshirtsize "Estimated Unisex T-Shirt Size (Next Level 3600)"

* Checks
tab tshirtsize
tab gender_num_cat tshirtsize, row // checks out with females having more
smaller sizes
count if missing(tshirtsize) //everything looks good

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v39.dta"


************************** SECTION 3 **********************************

******** Create a table with  statistics describing the anthropometric
measures of the entire sample

* generate table with mean, sd, min, and max and have it columns and 3
decimal points

tabstat ///
stature ///
heightcm ///
weightkg ///
weightkg_selfreport ///
```

```
cervicaleheight ///
waistheightomphalion ///
span ///
thumbtipreach ///
chestcircumference ///
waistcircumference ///
trochanterionheight ///
kneeheightmidpatella ///
functionalleglength ///
bicristalbreadth ///
hipbreadth ///
hipbreadthsitting ///
footlength ///
, statistics(mean sd min max) ///
columns(statistics) ///
format(%9.3f) ///
save

putexcel set "Anthropometric_Summary.xlsx", replace ///
putexcel A1 = matrix(r(StatTotal)'), names /// put into excel starting in
cell A1 and transpose for the correct layout // used ChatGPT to help identify
code

*** Create a table with the number of missing values and suspicious values
for each of the anthropometric measures listed above

misstable summarize /// check to see anthropometric variables that have
missing values
stature ///
heightcm ///
weightkg ///
weightkg_selfreport /// this is the only one that has a missing value. It has
1 observation missing.
cervicaleheight ///
waistheightomphalion ///
span ///
thumbtipreach ///
chestcircumference ///
waistcircumference ///
trochanterionheight ///
kneeheightmidpatella ///
functionalleglength ///
bicristalbreadth ///
hipbreadth ///
hipbreadthsitting ///
footlength

codebook /// double check that none are missing
stature ///
```

```
heightcm ///
weightkg ///
weightkg_selfreport ///
cervicaleheight ///
waistheightomphalion ///
span ///
thumbtipreach ///
chestcircumference ///
waistcircumference ///
trochanterionheight ///
kneeheightmidpatella ///
functionalleglength ///
bicristalbreadth ///
hipbreadth ///
hipbreadthsitting ///
footlength

** Get count for suspicious values (only need to look at flag variables
because that is the only way I identfied suspicious values)

count if flag_weightkg_selfreport > 0 // = 107
count if flag_heightcm > 0 // = 14
count if flag_footlength > 0 // = 36
count if flag_stature > 0 // = 15
count if flag_waistcircumference > 0 // = 125


******** Figure for the differences in % of total height attributable to the
height to hip by sex

* The formula for this figure is percent hip to height = trochanterion height
/ stature multiplied by 100. Trochanterion height is height to the hip.

** Create new variable

gen pct_height_to_hip = (trochanterionheight / stature) * 100 // create new
variable with formula describe above
label variable pct_height_to_hip "% of Total Height to Hip" // label

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v40.dta"

** Create figure

by gender_num_cat, sort: summarize pct_height_to_hip, detail // explore
descriptive stats of the new variable by gender

***  Box plot
graph box pct_height_to_hip, over(gender_num_cat) /// box plots by gender
```

```
asyvars /// allow for changing colors of each box/markers
box(1, fcolor(cranberry) lcolor(cranberry)) /// change colors
box(2, fcolor(navy) lcolor(navy)) ///
marker(1, mcolor(cranberry)) /// change colors
marker(2, mcolor(navy)) ///
plotregion(lcolor(black) lwidth(medium)) /// graph border and formatting
graphregion(color(white)) /// change background color
b1title(Gender, size(large) color(black) margin(t=1)) ///  adds formatted
title
ytitle("% of Total Height Attributable to Hip Height", size(medium)
margin(t=2)) ///  labels and formats y axis title
ylabel(45(2)57, format(%4.1f) labsize(large) tlwidth(medium) nogrid) /// sets
axis ticks, labels, and formatting; removes grid marks
name(fig3_1, replace) // names and saves


*** Bar chart

collapse /// collapse to mean, SD, and N by gender // used ChatGPT to help
identify code
    (mean) mean_pct = pct_height_to_hip ///
    (sd)   sd_pct   = pct_height_to_hip ///
    (count) n_pct   = pct_height_to_hip, ///
    by(gender_num_cat)

gen upper = mean_pct + sd_pct
gen lower = mean_pct - sd_pct

twoway /// get bar graph
(bar mean_pct gender_num_cat if gender_num_cat==1, /// add female
barwidth(.8) fcolor(cranberry) lcolor(cranberry)) /// formatting
(bar mean_pct gender_num_cat if gender_num_cat==2, /// add male
barwidth(.8) fcolor(navy) lcolor(navy)) /// formatting
(rcap upper lower gender_num_cat, /// add bars
lcolor(black) lwidth(medium)), /// formatting
xlabel(1 "Female" 2 "Male", nogrid) /// formatting
ytitle("% of Total Height Attributable to Hip Height") /// formatting
ylabel(49.5(0.5)53.5, format(%4.1f) nogrid) /// formatting
graphregion(color(white)) /// formatting
plotregion(lcolor(black)) /// formatting
name(fig_3_2, replace)

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project 1/Datasets/Figre
3.2.dta"
file /Users/savannahdaines/Desktop/DMAP/Assignments/Project 1/Datasets/Figre
3.2.dta saved
```

```
************************** SECTION 4 *********************************

 use "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v40.dta"
(ANSUR II exploratory data analysis dataset)

**** Correlation between measures of stature

****** Analysis

corr ///
stature ///
kneeheightmidpatella ///
cervicaleheight ///
trochanterionheight ///
waistheightomphalion ///
functionalleglength ///
footlength ///
thumbtipreach ///
span


             |  stature kneehe~a cervic~t trocha~t waisth~n functi~h footle~h
thumbt~h
-------------+------------------------------------------------------------
---------
     stature |   1.0000
kneeheight~a |   0.8894   1.0000
cervicaleh~t |   0.9912   0.9030   1.0000
trochanter~t |   0.8772   0.9255   0.8880   1.0000
waistheigh~n |   0.9367   0.9134   0.9414   0.9118   1.0000
functional~h |   0.8877   0.8759   0.9022   0.8655   0.8787   1.0000
  footlength |   0.8448   0.8077   0.8503   0.7695   0.8068   0.8003   1.0000
thumbtipre~h |   0.8060   0.7945   0.8250   0.7771   0.7815   0.8320   0.7873
1.0000
        span |   0.8988   0.8732   0.9079   0.8594   0.8835   0.8720   0.8627
0.8811

             |     span
-------------+---------
        span |   1.0000


*The two highest correlated are 1) stature & cervicalheight and 2) stature &
waistheightomphalion


********  Figures
```

```
* 1) stature & cervicalheight

twoway ///
(scatter stature cervicaleheight, /// do a scatterplot
msymbol(circle) /// formatting
msize(vsmall) /// formatting
mcolor(teal%10)) /// formatting
(lfit stature cervicaleheight, /// add best fit line
lcolor(black) /// formatting
lwidth(medium)), /// formatting
xscale(range(115 175)) /// formatting
xlabel(115(10)175, format(%4.0f) labsize(medium) tlwidth(medium) nogrid) ///
yscale(range(140 200)) /// formatting
ylabel(140(10)200, format(%4.0f) labsize(medium) tlwidth(medium) nogrid) ///
xtitle("Cervicale Height (cm)") /// formatting
ytitle("Stature (cm)") /// formatting
plotregion(lcolor(black) lwidth(medium)) /// formatting
graphregion(color(white) lcolor(black) lwidth(medium)) /// formatting
legend (off) /// formatting
name(fig_4_1, replace)

pwcorr stature cervicaleheight, sig // get p value // used ChatGPT to help
identify code

** add r with GUI r=0.991 and also change color of best fit line with GUI

*** Dealing with overplotting - hexplot creation

hexplot stature cervicaleheight, ///
bins(55) ///
color(viridis) ///
xlabel(115(10)175, labsize(medium) tlwidth(medium) nogrid) ///
ylabel(140(10)200, format(%4.0f) labsize(medium) tlwidth(medium) nogrid) ///
levels(7) ///
plotregion(lcolor(black) lwidth(medium)) ///
graphregion(color(white) lcolor(black) lwidth(medium)) ///
name(fig_4_2, replace)


**** Correlation between stature/other measures by sex

** full sample

corr ///  corrlation for anthro stature measures
stature ///
kneeheightmidpatella ///
cervicaleheight ///
trochanterionheight ///
waistheightomphalion ///
```

```
functionalleglength ///
footlength ///
thumbtipreach ///
span

     |  stature kneehe~a cervic~t trocha~t waisth~n functi~h footle~h
thumbt~h     span
-------------+------------------------------------------------------------
------------------
     stature |   1.0000
kneeheight~a |   0.8894   1.0000
cervicaleh~t |   0.9912   0.9030   1.0000
trochanter~t |   0.8772   0.9255   0.8880   1.0000
waistheigh~n |   0.9367   0.9134   0.9414   0.9118   1.0000
functional~h |   0.8877   0.8759   0.9022   0.8655   0.8787   1.0000
  footlength |   0.8448   0.8077   0.8503   0.7695   0.8068   0.8003   1.0000
thumbtipre~h |   0.8060   0.7945   0.8250   0.7771   0.7815   0.8320   0.7873
1.0000
        span |   0.8988   0.8732   0.9079   0.8594   0.8835   0.8720   0.8627
0.8811   1.0000



** female

corr ///  corrlation for anthro stature measures
stature ///
kneeheightmidpatella ///
cervicaleheight ///
trochanterionheight ///
waistheightomphalion ///
functionalleglength ///
footlength ///
thumbtipreach ///
span if gender_num_cat==1 // female


             |  stature kneehe~a cervic~t trocha~t waisth~n functi~h footle~h
thumbt~h
-------------+------------------------------------------------------------
---------
     stature |   1.0000
kneeheight~a |   0.8432   1.0000
cervicaleh~t |   0.9843   0.8664   1.0000
trochanter~t |   0.8578   0.9168   0.8786   1.0000
waistheigh~n |   0.9088   0.8802   0.9234   0.8897   1.0000
functional~h |   0.8323   0.8341   0.8497   0.8484   0.8289   1.0000
  footlength |   0.7219   0.7315   0.7262   0.7184   0.7016   0.7124   1.0000
```

```
thumbtipre~h |   0.6710   0.6853   0.6904   0.6935   0.6573   0.7338   0.6535
1.0000
        span |   0.8190   0.8319   0.8309   0.8423   0.8284   0.8132   0.7726
0.8064

             |     span
-------------+---------
        span |   1.0000
```

* the highest correlations are still between 1) stature & cervicalheight and 2) stature & waistheightomphalion

** Male

```
corr /// corrlation for anthro stature measures
stature ///
kneeheightmidpatella ///
cervicaleheight ///
trochanterionheight ///
waistheightomphalion ///
functionalleglength ///
footlength ///
thumbtipreach ///
span if gender_num_cat==2 // male

             |  stature kneehe~a cervic~t trocha~t waisth~n functi~h footle~h
thumbt~h
-------------+------------------------------------------------------------
---------
     stature |   1.0000
kneeheight~a |   0.8334   1.0000
cervicaleh~t |   0.9841   0.8572   1.0000
trochanter~t |   0.8501   0.8981   0.8698   1.0000
waistheigh~n |   0.9085   0.8701   0.9157   0.8843   1.0000
functional~h |   0.8179   0.8084   0.8458   0.8114   0.8125   1.0000
  footlength |   0.7181   0.6925   0.7253   0.6758   0.6931   0.6608   1.0000
thumbtipre~h |   0.6912   0.6981   0.7263   0.6997   0.6759   0.7462   0.6524
1.0000
        span |   0.8230   0.8000   0.8386   0.8135   0.8178   0.7894   0.7492
0.8119

             |     span
-------------+---------
        span |   1.0000
```

* the highest correlations are still between 1) stature & cervicalheight and 2) stature & waistheightomphalion

```
******* Compare the self-reported weight to the measured weight.

* Create difference variabel

*** want the difference to represent: Reported – Measured, so a negative
value is underreproted and positive value is overeported

gen diff_weight = weightkg_selfreport - weightkg // create

label variable diff_weight "Self-reported – Measured Weight (kg)" //label

summ diff_weight, detail // look at distribution

* Figures

** Box
graph box diff_weight, horizontal /// creates horizontal box plot
box(1, fcolor(teal) lcolor(teal)) /// changes box fill and ouline color
marker(1, mcolor(teal%55)) /// sets outlier/outside points color
plotregion(lstyle(box) lcolor(black) lwidth(large)) /// graph border and
formatting
graphregion(color(white)) /// chanes background color
ytitle("Weight Reporting Bias (Self-reported - Measured kg)") ///
ylabel(-50(10)70, format(%4.0f) labsize(large) tlwidth(medium) nogrid) //
Sets axis ticks, labels, and formatting; removes grid marks
name(fig_4_3, replace)

** Hexplot
hexplot diff_weight weightkg, ///
bins(50) ///
color(viridis) ///
xlabel(40(10)140, labsize(medium) tlwidth(medium) nogrid) ///
 ylabel(-50(10)70, format(%4.0f) labsize(medium) tlwidth(medium) nogrid) ///
ytitle("Weight Reporting Bias (Self-reported - Measured kg)") ///
plotregion(lcolor(black) lwidth(medium)) ///
graphregion(color(white) lcolor(black) lwidth(medium)) ///
aspectratio(1) ///
levels(7) ///
addplot( ///
function y = 0, ///
range(35 145) ///
lcolor(red) ///
lwidth(medium)) ///
name(fig_4_4, replace)
```

```
******* Extent to which the difference between reported and measured weight
varies by sex.

bysort gender_num_cat: summ diff_weight, detail
 // distriubtion // used ChatGPT to help identify code

-> gender_num_cat = Female

                        diff_weight
-------------------------------------------------------------
      Percentiles      Smallest
 1%    -8.575512      -43.40794
 5%    -5.037643      -15.89977
10%     -3.59819      -14.83765       Obs               1,985
25%     -1.92041      -13.99705       Sum of wgt.       1,985

50%    -.5950127                      Mean          -.8584035
                        Largest       Std. dev.      2.647028
75%     .5315208       7.296371
90%     1.685032       7.969391       Variance       7.006757
95%     2.559635       14.40295       Skewness      -2.657758
99%     4.607708       16.09319       Kurtosis       39.94133

-------------------------------------------------------------------------------
----------------
-> gender_num_cat = Male

                        diff_weight
-------------------------------------------------------------
      Percentiles      Smallest
 1%    -8.921318      -48.27007
 5%    -4.926758      -45.51338
10%     -3.29705      -44.68526       Obs               4,082
25%    -1.526299      -22.19116       Sum of wgt.       4,082

50%    -.0646286                      Mean          -.2365261
                        Largest       Std. dev.      3.190103
75%     1.329933       10.70295
90%     2.802948       11.30726       Variance       10.17675
95%     3.835373       17.90023       Skewness      -.4637308
99%     6.332649        66.1263       Kurtosis       81.85347


tabstat diff_weight, by(gender_num_cat) stat(mean sd) // get summary table
with mean and sd // used ChatGPT to help identify code


gender_num_cat |      Mean        SD
---------------+--------------------
```

```
        Female | -.8584035  2.647028
          Male | -.2365261  3.190103
---------------+--------------------
         Total | -.4399918  3.037008
------------------------------------

**create bar chart

collapse /// collapse to mean, SD, and N by gender // used ChatGPT to help
identify code
    (mean) mean_diff_weight = diff_weight ///
    (sd)   sd_diff_weight = diff_weight ///
    (count) n_diff_weight = diff_weight, ///
    by(gender_num_cat)

gen upper = mean_diff_weight + sd_diff_weight // used ChatGPT to help
identify code
gen lower = mean_diff_weight - sd_diff_weight // used ChatGPT to help
identify code

twoway /// get bar graph
(bar mean_diff_weight gender_num_cat if gender_num_cat==1, /// females
barwidth(.8) fcolor(cranberry) lcolor(cranberry)) /// formatting
(bar mean_diff_weight gender_num_cat if gender_num_cat==2, /// males
barwidth(.8) fcolor(navy) lcolor(navy)) /// formatting
(rcap upper lower gender_num_cat, /// // used ChatGPT to help identify code
lcolor(black) lwidth(medium)), ///    formatting
xlabel(1 "Female" 2 "Male", nogrid noticks) /// formatting
ytitle("Weight Reporting Bias (Self-reported - Measured kg)") /// formatting
ylabel(-3.5(1)3.5, format(%4.1f) nogrid) /// formatting
yline(0, lpattern(shortdash) lcolor(lightgrey) lwidth(thin)) /// formatting
graphregion(color(white)) /// formatting
plotregion(lcolor(black)) /// formatting
name(fig_4_5, replace)

** explore bias in more depth

ttest diff_weight, by(gender_num_cat) // t test for signficance

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf.
interval]
---------+--------------------------------------------------------------
  Female |   1,985   -.8584035    .0594126    2.647028   -.9749211   -
.7418859
```

```
    Male |   4,082   -.2365261    .0499308    3.190103   -.3344176   -
.1386345
---------+--------------------------------------------------------------
-
Combined |   6,067   -.4399918    .0389905    3.037008   -.5164271   -
.3635566
---------+--------------------------------------------------------------
-
    diff |           -.6218774    .0827252               -.7840483   -
.4597066
------------------------------------------------------------------------------
-
    diff = mean(Female) - mean(Male)                              t =  -
7.5174
H0: diff = 0                                     Degrees of freedom =
6065

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) =
1.0000


reg diff_weight i.gender_num_cat // used ChatGPT to help identify code

   Source |       SS           df       MS      Number of obs   =     6,067
-------------+----------------------------------   F(1, 6065)      =
56.51
       Model |   516.49856         1   516.49856   Prob > F        =
0.0000
    Residual |  55432.7427     6,065  9.13977621   R-squared       =
0.0092
-------------+----------------------------------   Adj R-squared   =
0.0091
       Total |  55949.2413     6,066  9.22341597   Root MSE        =
3.0232

------------------------------------------------------------------------------
---
   diff_weight | Coefficient  Std. err.      t    P>|t|     [95% conf.
interval]
---------------+--------------------------------------------------------------
---
gender_num_cat |
         Male  |   .6218774   .0827252     7.52   0.000     .4597066
.7840483
         _cons |  -.8584035   .0678559   -12.65   0.000    -.9914251   -
.7253819
```

```
* Odds ratio

gen underreport = diff_weight < 0 // create variable where 1 = underreported,
0 accurate or overreported

label variable underreport "Underreported Weight (Self-reported < Measured)"
// label
label define underlbl 0 "No" 1 "Yes"
label values underreport underlbl
tab underreport // check labels

tab underreport gender_num_cat, col // check

logistic underreport i.gender_num_cat // get odds ratio / i means treat like
a categorical variable // used ChatGPT to help identify code

Logistic regression                                     Number of obs =
6,068
                                                        LR chi2(1)    =
89.94
                                                        Prob > chi2   =
0.0000
Log likelihood = -4122.8655                             Pseudo R2     =
0.0108

------------------------------------------------------------------------------
   underreport | Odds ratio   Std. err.      z    P>|z|     [95% conf.
interval]
---------------+--------------------------------------------------------------
gender_num_cat |
         Male  |   .5891406    .033187    -9.39   0.000     .5275573
.6579126
         _cons |   1.795775   .0840907    12.50   0.000     1.638297
1.968389
------------------------------------------------------------------------------
Note: _cons estimates baseline odds.

// so OR (male) 0.589, 95% CI 0.528 - 0.658, p < 0.001 // so males less
likely to underreport


******* Compare the self-reported height to the measured height.

* Create difference variabel
```

```
*** want the difference to represent: Reported – Measured, so a negative
value is underreproted and positive value is overvierported

gen diff_height = heightcm - stature // create  variable

label variable diff_height "Self-Reported Height – Stature (cm)" // label

summ diff_height, detail // look at distribution

** Figures

* Box
graph box diff_height, horizontal /// creates horizontal box plot
box(1, fcolor(teal) lcolor(teal)) /// changes box fill and ouline color
marker(1, mcolor(teal%55)) /// sets outlier/outside points color
plotregion(lstyle(box) lcolor(black) lwidth(large)) /// graph border and
formatting
graphregion(color(white)) /// chanes background color
ytitle("Height Reporting Bias (Self-reported - Measured cm)") ///
ylabel(-30(10)70, format(%4.0f) labsize(large) tlwidth(medium) nogrid) //
Sets axis ticks, labels, and formatting; removes grid marks
name(fig_4_6, replace)

* Hex plot
hexplot diff_weight stature, ///
bins(50) ///
color(viridis) ///
xlabel(140(10)200, labsize(medium) tlwidth(medium) nogrid) ///
 ylabel(-50(10)70, format(%4.0f) labsize(medium) tlwidth(medium) nogrid) ///
ytitle("Height Reporting Bias (Self-reported - Measured cm)") ///
xtitle("Measured Height (cm)") ///
plotregion(lcolor(black) lwidth(medium)) ///
graphregion(color(white) lcolor(black) lwidth(medium)) ///
aspectratio(1) ///
levels(7) ///
addplot( /// used ChatGPT to help identify code
function y = 0, ///
range(140 200) ///
lcolor(red) ///
lwidth(medium)) ///
name(fig_4_7, replace)

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v41.dta"


******** Relationship between derived `body type' and weight/BMI
```

```
* a) Compare the distributions of weight and BMI for each of the body types
you created in 1e above. Describe your findings in a paragraph with a figure
and/or table. Focus on how well your body types describe differences in
weight and obesity.*

*** distributions
tabstat weightkg bmi, by(bodytype) stat(mean sd) // exploratory

graph box bmi, over(bodytype) // exploratory
graph box weightkg, over(bodytype) // exploratory

tabstat weightkg bmi, by(bodytype) stat(mean sd p25 p50 p75 n) // exploratory
// used ChatGPT to help identify code

**Figure for weight
vioplot weightkg, over(bodytype) ///
ytitle("Weight (kg)") ///
graphregion(color(white)) ///
plotregion(lcolor(black) lwidth(thin)) ///
graphregion(color(white) lcolor(black) lwidth(thin)) ///
ylabel(40(20)140, format(%4.0f) labsize(medium) tlwidth(thin) nogrid) ///
xlabel(, angle(45) labgap(2) labsize(small) tlwidth(thin) nogrid) /// used
ChatGPT to help identify code
xsize(80) ysize(60) ///
bar(color(black)) density(color(teal%50)) line(color(teal))
median(color(red)) ///
name(fig_4_8, replace)


**Figure for bmi continous
vioplot bmi, over(bodytype) ///
ytitle("Body Mass Index (BMI)") ///
graphregion(color(white)) ///
plotregion(lcolor(black) lwidth(thin)) ///
graphregion(color(white) lcolor(black) lwidth(thin)) ///
ylabel(15(5)45, format(%4.0f) labsize(medium) tlwidth(thin) nogrid) ///
xlabel(, angle(45) labgap(2) labsize(small) tlwidth(thin) nogrid) /// used
ChatGPT to help identify code
xsize(80) ysize(60) ///
bar(color(black)) density(color(teal%50)) line(color(teal))
median(color(red)) ///
name(fig_4_9, replace)


**Differences for BMI categorical /obesity

tab bodytype bmi_cat, row nofreq // get table; used ChatGPT to help identify
code
```

```
************************** SECTION 5 **********************************


** Characterize the association of BMI and age (continuous variables),
stratified by sex. Present with figure and table

* Table

reg bmi age if gender_num_cat == 1 // female regression

      Source |       SS           df       MS      Number of obs   =
1,985
-------------+----------------------------------   F(1, 1983)      =
137.57
       Model |  1569.98189         1  1569.98189   Prob > F        =
0.0000
    Residual |  22630.6643     1,983   11.412337   R-squared       =
0.0649
-------------+----------------------------------   Adj R-squared   =
0.0644
       Total |  24200.6462     1,984  12.1979064   Root MSE        =
3.3782

------------------------------------------------------------------------------
         bmi | Coefficient  Std. err.      t    P>|t|     [95% conf.
interval]
-------------+----------------------------------------------------------------
         age |    .106792    .009105    11.73   0.000     .0889357
.1246483
       _cons |   22.40447   .2742748    81.69   0.000     21.86657
22.94237
------------------------------------------------------------------------------


reg bmi age if gender_num_cat == 2 // male regression

     Source |       SS           df       MS      Number of obs   =     4,083
-------------+----------------------------------   F(1, 4081)      =
333.46
       Model |  5032.80109         1  5032.80109   Prob > F        =
0.0000
    Residual |  61593.2659     4,081  15.0926895   R-squared       =
0.0755
-------------+----------------------------------   Adj R-squared   =
0.0753
```

```
       Total |   66626.067     4,082  16.3219174   Root MSE        =
3.8849

------------------------------------------------------------------------------
         bmi | Coefficient  Std. err.      t    P>|t|     [95% conf.
interval]
-------------+----------------------------------------------------------------
         age |   .1260303   .0069017    18.26   0.000     .1124992
.1395613
       _cons |   23.88325   .2168704   110.13   0.000     23.45807
24.30844
------------------------------------------------------------------------------


* Figure

** Option 1
twoway ///
(scatter bmi age if gender_num_cat==1, ///
      msymbol(circle) msize(vsmall) mcolor(cranberry%10)) ///
      (lfit bmi age if gender_num_cat==1, lcolor(cranberry)) /// female;
used ChatGPT to help identify code
(scatter bmi age if gender_num_cat==2, /// male
      msymbol(circle) msize(vsmall) mcolor(navy%10)) ///
      (lfit bmi age if gender_num_cat==2, lcolor(navy)), /// used ChatGPT to
help identify code
by(gender_num_cat, col(2) note("") ixaxes iyaxes) /// used ChatGPT to help
identify code
xtitle("Age (years)") ///
xscale(range(18 60)) /// adjusts x axis to 0-75
xlabel(18(4)60, grid glcolor(white)labsize(medium) tlwidth(medium) nogrid)
/// sets tick marks; start(unit increase)end, also formats
ylabel(15(5)45, format(%4.0f)labsize(medium) tlwidth(medium) nogrid) /// sets
tick marks and formats labels
ytitle("BMI (kg/m²)") ///
xsize(12) ysize(5) ///
plotregion(lcolor(black) lwidth(medium)) ///
graphregion(color(white) lcolor(black) lwidth(medium)) ///
legend (off) ///
name(fig5_1, replace)

** Option 2 (size differences)
twoway ///  used ChatGPT to help identify code
(scatter bmi age if gender_num_cat==1, ///
      msymbol(circle) msize(vsmall) mcolor(cranberry%10)) ///
      (lfit bmi age if gender_num_cat==1, lcolor(cranberry)) ///
```

```
(scatter bmi age if gender_num_cat==2, ///
      msymbol(circle) msize(vsmall) mcolor(navy%10)) ///
      (lfit bmi age if gender_num_cat==2, lcolor(navy)), ///
by(gender_num_cat, col(1) note("") ixaxes iyaxes) ///
xtitle("Age (years)") ///
xscale(range(18 60)) /// adjusts x axis to 0-75
xlabel(18(4)60, labsize(medium) tlwidth(medium) nogrid noticks) /// sets tick
marks; start(unit increase)end, also formats
ylabel(15(5)45, format(%4.0f)labsize(medium) tlwidth(medium) nogrid noticks)
/// sets tick marks and formats labels
ytitle("BMI (kg/m²)") ///
xsize(10) ysize(15) ///
plotregion(lcolor(black) lwidth(medium)) ///
graphregion(color(white) lcolor(black) lwidth(medium)) ///
name(fig5_1a, replace)

** Get r and p values to add through the GUI
pwcorr bmi age if gender_num_cat==1, sig // get r and p value female
           |      bmi      age
-------------+------------------
         bmi |   1.0000
             |
             |
         age |   0.2547   1.0000
             |   0.0000


pwcorr bmi age if gender_num_cat==2, sig // get r and p value male


             |      bmi      age
-------------+------------------
         bmi |   1.0000
             |
             |
         age |   0.2748   1.0000
             |   0.0000



******** Characterize the association of BMI and age, stratified by sex,
using categorical measures BM and age.

*** Variable creation
gen age_num_cat = . // create numerical categorical variable for age
replace age_num_cat = 1 if age >=18 & age <=24 // Use NCHS standard age
categories
replace age_num_cat = 2 if age >=25 & age <=44
replace age_num_cat = 3 if age >=45 & age <=64
```

```
replace age_num_cat = 4 if age >=65

label define age_num_catlbl 1 "18-24 years" 2 "25-44" 3 "45-64" 4 "65+" //
label
label values age_num_cat age_num_catlbl
label variable age_num_cat "NCHS standard age categories"

tab age_num_cat // check

save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v42.dta"

*** Table
tab bmi_cat age_num_cat if gender_num_cat==1, col chi2 // get frequencies and
count and chi square for females // used ChatGPT to help identify code

tab bmi_cat age_num_cat if gender_num_cat==2, col chi2 // get frequencies and
count and chi square for males // used ChatGPT to help identify code


tab bmi_cat age_num_cat if gender_num_cat==1, chi2 V // get cramer's v //
used ChatGPT to help identify code

tab bmi_cat age_num_cat if gender_num_cat==2, chi2 V // get cramer's v //
used ChatGPT to help identify code

*** Figure

graph bar (count), /// bar char
over(bmi_cat) /// break it down by bmi category
over(age_num_cat) /// include a physi
stack asyvars /// make it stacked
percentages /// show percentages
by(gender_num_cat, col(1)) /// include male and female, 1 column format
bar(1, color(stc1)) /// adjust colors of bmi categories
bar(2, color(stc3)) ///
bar(3, color(stc4)) ///
bar(4, color(stc2)) ///
blabel(bar, color(black) format(%6.2f) suffix("%") position(center) size
(medsmall)) /// adjust percent formatting in bars // used ChatGPT to help
identify code
ytitle(Percent) /// title y axis
ylabel(0(25)100, nogrid) /// change y axis ticks and remove gridlines
plotregion(lcolor(black) lwidth(thin)) /// add border
name(fig5_2, replace) // names and saves

*** Missing labels:

label variable obese "Not obese or obese"
```

```
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project
1/Datasets/ansur2allV2_v43.dta"

*** Final dataset:
save "/Users/savannahdaines/Desktop/DMAP/Assignments/Project 1/Project 1
Submission/ansur2allV2_finalsubmission.dta
```